CHAPTER 9

PARTIAL DIFFERENTIAL EQUATIONS

9.1 INTRODUCTION

As mentioned in Chapter 7, partial differential equations (PDEs) involve derivatives with respect to more than one independent variable; if the independent variables are x and y, a PDE in a dependent variable $\varphi(x, y)$ will contain **partial derivatives**, with the meaning discussed in Eq. (1.141). Thus, $\partial\varphi/\partial x$ implies an x derivative with y held constant, $\partial^2\varphi/\partial x^2$ is the second derivative with respect to x (again keeping y constant), and we may also have **mixed derivatives**

$$\frac{\partial^2\varphi}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial\varphi}{\partial y}\right).$$

Like ordinary derivatives, partial derivatives (of any order, including mixed derivatives) are linear operators, since they satisfy equations of the type

$$\frac{\partial[\varphi(x, y) + b\varphi(x, y)]}{\partial x} = a\frac{\partial\varphi(x, y)}{\partial x} + b\frac{\partial\varphi(x, y)}{\partial x}.$$

Similar to the situation for ODEs, general differential operators, $\mathcal{L}$, which may contain partial derivatives of any order, pure or mixed, multiplied by arbitrary functions of the independent variables, are **linear** operators, and equations of the form

$$\mathcal{L}\varphi(x, y) = F(x, y)$$

are linear PDEs. If the **source term** $F(x, y)$ vanishes, the PDE is termed **homogeneous**; if $F(x, y)$ is nonzero, it is **inhomogeneous**.

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Homogeneous PDEs have the property, previously noted in other contexts, that any linear combination of solutions will also be a solution to the PDE. This is the **superposition principle** that is fundamental in electrodynamics and quantum mechanics, and which also permits us to build specific solutions by the linear combination of suitable members of the set of functions constituting the general solution to the homogeneous PDE.

Example 9.1.1 Various Types of PDEs

Laplace	$\nabla^2\psi = 0,$	linear, homogeneous
Poisson	$\nabla^2\psi = f(\mathbf{r}),$	linear, inhomogeneous
Euler (inviscid flow)	$\dfrac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u} = -\dfrac{\nabla P}{\rho}$	nonlinear, inhomogeneous

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Since the dynamics of many physical systems involve just two derivatives, for example, acceleration in classical mechanics, and the kinetic energy operator $\sim\nabla^2$ in quantum mechanics, differential equations of second order occur most frequently in physics. Even when the defining equations are first order, they may, as in Maxwell's equations, involve two coupled unknown vector functions (they are the electric and magnetic fields), and the elimination of one unknown vector yields a second-order PDE for the other (compare Example 3.6.2).

Examples of PDEs

Among the most frequently encountered PDEs are the following:

1. Laplace's equation, $\nabla^2\psi = 0$.
   This very common and very important equation occurs in studies of
   (a) electromagnetic phenomena, including electrostatics, dielectrics, steady currents, and magnetostatics,
   (b) hydrodynamics (irrotational flow of perfect fluid and surface waves),
   (c) heat flow,
   (d) gravitation.
2. Poisson's equation, $\nabla^2\psi = -\rho/\varepsilon_0$.
   This inhomogeneous equation describes electrostatics with a source term $-\rho/\varepsilon_0$.
3. Helmholtz and time-independent diffusion equations, $\nabla^2\psi \pm k^2\psi = 0$.
   These equations appear in such diverse phenomena as
   (a) elastic waves in solids, including vibrating strings, bars, membranes,
   (b) acoustics (sound waves),
   (c) electromagnetic waves,
   (d) nuclear reactors.

4. The time-dependent diffusion equation, $\nabla^2\psi = \dfrac{1}{a^2}\dfrac{\partial\psi}{\partial t}$.
5. The time-dependent classical wave equation, $\dfrac{1}{c^2}\dfrac{\partial^2\psi}{\partial t^2} = \nabla^2\psi$.
6. The Klein-Gordon equation, $\partial^2\psi = -\mu^2\psi$, and the corresponding vector equations in which the scalar function ψ is replaced by a vector function. Other, more complicated forms are also common.
7. The time-dependent Schrödinger wave equation,

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t}$$

   and its time-independent form

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi.$$

8. The equations for elastic waves and viscous fluids and the telegraphy equation.
9. Maxwell's coupled partial differential equations for electric and magnetic fields and those of Dirac for relativistic electron wave functions.

We begin our study of PDEs by considering first-order equations, which illustrate some of the most important principles involved. We then continue to classification and properties of second-order PDEs, and a preliminary discussion of prototypical homogeneous equations of the different classes. Finally, we examine a very useful and powerful method for obtaining solutions to homogeneous PDEs, namely the method of **separation of variables**.

This chapter is mainly devoted to general properties of homogeneous PDEs; full detail on specific equations is for the most part postponed to chapters that discuss the special functions involved. Questions arising from the extension to inhomogeneous PDEs (i.e., problems involving **sources** or **driving terms**) are also deferred, mainly to later chapters on **Green's functions** and integral transforms.

Occasionally, we encounter equations of higher order. In both the theory of the slow motion of a viscous fluid and the theory of an elastic body we find the equation

$$(\nabla^2)^2\psi = 0.$$

Fortunately, these higher-order differential equations are relatively rare and are not discussed here. Sometimes, particularly in fluid mechanics, we encounter nonlinear PDEs.

9.2 First-Order Equations

While the most important PDEs arising in physics are linear and second order, many involving three spatial variables plus possibly a time variable, first-order PDEs do arise (e.g., the Cauchy-Riemann equations of complex variable theory). Part of the motivation for studying these easily solved equations is that the study provides insights that apply also to higher-order problems.

Characteristics

Let us start by considering the following homogeneous linear first-order equation in two independent variables x and y, with constant coefficients a and b, and with dependent variable $\varphi(x,y)$:

$$\mathcal{L}\varphi = a\frac{\partial\varphi}{\partial x} + b\frac{\partial\varphi}{\partial y} = 0. \tag{9.1}$$

This equation would be easier to solve if we could rearrange it so that it contained only one derivative; one way to do this is would be to rewrite our PDE in terms of new coordinates (s,t) such that one of them, say s, is such that $(\partial/\partial s)_t$ would expand into the linear combination of $\partial/\partial x$ and $\partial/\partial y$ in the original PDE, while the other new coordinate, t, is such that $(\partial/\partial t)_s$ does not occur in the PDE. It is easily verified that definitions of s and t consistent with these objectives for the PDE in Eq. (9.1) are $s = ax + by$ and $t = bx - ay$. To check this, write $\varphi(x,y) = \varphi(x(s,t), y(s,t)) = \hat{\varphi}(s,t)$, and we can verify that

$$\left(\frac{\partial\varphi}{\partial x}\right)_y = a\left(\frac{\partial\varphi}{\partial s}\right)_t + b\left(\frac{\partial\varphi}{\partial t}\right)_s \quad \text{and} \quad \left(\frac{\partial\varphi}{\partial y}\right)_x = b\left(\frac{\partial\varphi}{\partial s}\right)_t - a\left(\frac{\partial\varphi}{\partial t}\right)_s,$$

so

$$a\frac{\partial\varphi}{\partial x} + b\frac{\partial\varphi}{\partial y} = (a^2+b^2)\frac{\partial\hat{\varphi}}{\partial s}.$$

We see that the PDE does not contain a derivative with respect to t. Since our PDE now has the simple form

$$(a^2+b^2)\frac{\partial\hat{\varphi}}{\partial s} = 0,$$

it clearly has solution

$$\hat{\varphi}(s,t) = f(t), \quad \text{with } f(t) \text{ completely arbitrary.} \tag{9.2}$$

In terms of the original variables,

$$\varphi(x,y) = f(bx - ay), \tag{9.3}$$

where we again stress that $f(t)$ is an arbitrary function of its argument.

Checking our work to this point, we note that

$$\mathcal{L}\varphi = a\frac{\partial f(bx-ay)}{\partial x} + b\frac{\partial f(bx-ay)}{\partial y} = abf'(bx-ay) + b\left[-af'(bx-ay)\right] = 0.$$

Since the satisfaction of this equation does not depend on the properties of the function f, we verify that $\varphi(x,y)$ as given in Eq. (9.3) is a solution of our PDE, **irrespective of the choice of the function** f. In fact, it is the general solution of our PDE.

It is useful to visualize the significance of what we have just observed. Note that holding $t = bx - ay$ to a fixed value defines a line in the xy plane on which our solution φ is constant, with individual points on this line corresponding to different values of $s = ax + by$. In addition, we observe that the lines of constant s are orthogonal to those of constant t, and that s has the same coefficients as the derivatives in the PDE. The general solution to our PDE can thus be characterized as **independent of s and with arbitrary dependence on t**.

The curves of constant t are called **characteristic curves**, or more frequently just **characteristics** of our PDE. An alternative and insightful way of describing the characteristic curves is to observe that they are the stream lines (flow lines) of s. Put another way, they are the lines that are traced out as the value of s is changed, keeping t constant. The characteristic can also be characterized by its slope,

$$\frac{dy}{dx} = \frac{b}{a}, \quad \text{for } \mathcal{L} \text{ in Eq. (9.1).} \tag{9.4}$$

For our present first-order PDE, the solution φ is constant along each characteristic. We shall shortly see that more general PDEs can be solved using ODE methods on characteristic lines, a feature that causes it to be said that PDE solutions **propagate** along the characteristics, giving further significance to the notion that in some sense these are lines of flow. In the present problem this translates into the statement that if we know φ at any point on a characteristic, we know it on the entire characteristic line.

The characteristics have one additional (but related) property of importance. Ordinarily, if a PDE solution $\varphi(x, y)$ is specified on a curve segment (a **boundary condition**), one can deduce from it the values of the solution at nearby points that are not on the curve. If one introduces a Taylor expansion about some point (x_0, y_0) on the curve (thereby tacitly assuming that there are no singularities that invalidate the expansion), the value of φ at a nearby point (x, y) will be given by

$$\varphi(x, y) = \varphi(x_0, y_0) + \frac{\partial\varphi(x_0, y_0)}{\partial x}(x - x_0) + \frac{\partial\varphi(x_0, y_0)}{\partial y}(y - y_0) + \cdots . \tag{9.5}$$

To use Eq. (9.5), we need values of the derivatives of φ. To obtain these derivatives, note the following:

- The specification of φ on a given curve, with the curve parametrically described by $x(l), y(l)$, means that the curve direction, i.e., dx/dl and dy/dl, is known, as is the derivative of φ along the curve, namely
$$\frac{d\varphi}{dl} = \frac{\partial\varphi}{\partial x}\frac{dx}{dl} + \frac{\partial\varphi}{\partial y}\frac{dy}{dl}. \tag{9.6}$$
Equation (9.6) therefore provides us with a linear equation satisfied by the two derivatives $\partial\varphi/\partial x$ and $\partial\varphi/\partial y$.
- The PDE supplies a second linear equation, in this case
$$a\frac{\partial\varphi}{\partial x} + b\frac{\partial\varphi}{\partial y} = 0. \tag{9.7}$$
- Providing that the determinant of their coefficients is not zero, we can solve Eqs. (9.6) and (9.7) for $\partial\varphi/\partial x$ and $\partial\varphi/\partial y$ at (x_0, y_0) and therefore evaluate the leading terms of the Taylor series for $\varphi(x, y)$.[1] The determinant of coefficients of Eqs. (9.6) and (9.7) takes the form
$$D = \begin{vmatrix} \dfrac{dx}{dl} & \dfrac{dy}{dl} \\ a & b \end{vmatrix} = b\frac{dx}{dl} - a\frac{dy}{dl}.$$

[1] The linear terms are all that are necessary; one can choose x and y close enough to (x_0, y_0) that second- and higher-order terms can be made negligible relative to those retained.

Now we make the observation that if φ was specified along a characteristic (for which $t = bx - ay$ =constant), we have

$$b\,dx - a\,dy = 0, \quad \text{or} \quad b\frac{dx}{dl} - a\frac{dy}{dl} = 0,$$

so that $D = 0$ and we cannot solve for the derivatives of φ. Our conclusions relative to characteristics, which can be extended to more general equations, are:

1. *If the dependent variable φ of the PDE in Eq. (9.1) is specified along a curve (i.e., φ has a* **boundary condition** *specified on a* **boundary curve***), this fixes the value of φ at a point of each characteristic that intersects the boundary curve, and hence at all points of each such characteristic;*
2. *If the boundary curve is along a characteristic, the boundary condition on it will ordinarily lead to inconsistency, and therefore, unless the boundary condition is redundant (i.e., coincidentally equal everywhere to the solution constructed from the value of φ at any one point on the characteristic), the PDE will not have a solution;*
3. *If the boundary curve has more than one intersection with the same characteristic, this will usually lead to an inconsistency, as the PDE may not have a solution that is simultaneously consistent with the values of φ at both intersections; and*
4. *Only if the boundary curve is* **not** *a characteristic can a boundary condition fix the value of φ at points not on the curve. Values of φ specified only on a characteristic of the PDE provide no information as to the value of φ at points not on that characteristic.*

In the above example, the argument t of the arbitrary function f was a linear combination of x and y, which worked because the coefficients of the derivatives in the PDE were constants. If these coefficients were more general functions of x and y, the foregoing type of analysis could still be carried out, but the form of t would have to be different. This more complicated case is illustrated in Exercises 9.2.5 and 9.2.6.

More General PDEs

Consider now a first-order PDE of a form more general than Eq. (9.1),

$$\mathcal{L}\varphi = a\frac{\partial\varphi}{\partial x} + b\frac{\partial\varphi}{\partial y} + q(x, y)\varphi = F(x, y). \tag{9.8}$$

We may identify its characteristic curves just as before, which amounts to making a transformation to new variables $s = ax + by$, $t = bx - ay$, in terms of which our PDE becomes, compare Eq. (9.5),

$$(a^2 + b^2)\left(\frac{\partial\varphi}{\partial s}\right) + \hat{q}(s, t)\hat{\varphi} = \hat{F}(s, t). \tag{9.9}$$

Here $\hat{q}(s, t)$ is obtained by converting $q(x, y)$ to the new coordinates:

$$\hat{q}(s, t) = q\left(\frac{as + bt}{a^2 + b^2}, \frac{bs - at}{a^2 + b^2}\right),$$

and $\hat{F}$ is related in a similar fashion to F. Equation (9.9) is really an ODE in s (containing what can be viewed as a parameter, t), and its general solution can be obtained by the usual procedures for solving ODEs.

Example 9.2.1 ANOTHER FIRST-ORDER PDE

Consider the PDE

$$\frac{\partial\varphi}{\partial x} + \frac{\partial\varphi}{\partial y} + (x+y)\varphi = 0.$$

Applying a transformation to the characteristic direction $t = x - y$ and the direction orthogonal thereto $s = x + y$, our PDE becomes

$$2\frac{\partial\varphi}{\partial s} + s\varphi = 0.$$

This equation separates into

$$2\frac{d\varphi}{\varphi} + s\,ds = 0,$$

with general solution

$$\ln\varphi = -\frac{s^2}{4} + C(t), \quad \text{or} \quad \varphi = e^{-s^2/4} f(t),$$

where $f(t)$, originally $\exp[C(t)]$, is completely arbitrary. One can simplify the result slightly by noting that $s^2/4 = t^2/4 + xy$; then $\exp(-t^2/4)$ can be absorbed into $f(t)$, leaving the compact result (in terms of x and y)

$$\varphi(x, y) = e^{-xy} f(x - y), \quad (f \text{ arbitrary}).$$

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More Than Two Independent Variables

It is useful to consider how the concept of characteristic can be generalized to PDEs with more than two independent variables. Given the three-dimensional (3-D) differential form

$$a\frac{\partial\varphi}{\partial x} + b\frac{\partial\varphi}{\partial y} + c\frac{\partial\varphi}{\partial z},$$

we apply a transformation to convert our PDE to the new variables $s = ax + by + cz$, $t = \alpha_1 x + \alpha_2 y + \alpha_3 z$, $u = \beta_1 x + \beta_2 y + \beta_3 z$, with α_i and β_i such that (s, t, u) form an orthogonal coordinate system. Then our 3-D differential form is found equivalent to

$$(a^2 + b^2 + c^2)\frac{\partial\varphi}{\partial s},$$

and the stream lines of s (those with t and u constant) are our characteristics, along which we can propagate a solution φ by solving an ODE. Each characteristic can be identified by its fixed values of t and u. For the 3-D analog of Eq. (9.1),

$$a\frac{\partial\varphi}{\partial x}+b\frac{\partial\varphi}{\partial y}+c\frac{\partial\varphi}{\partial z}=0, \tag{9.10}$$

we have

$$(a^2+b^2+c^2)\frac{\partial\varphi}{\partial s}=0,$$

with solution $\varphi=f(t,u)$, with f a completely arbitrary function of its two arguments.

Consider next an attempt to solve our 3-D PDE subject to a boundary condition fixing the values of the PDE solution φ on a surface. If the characteristic through a point on the surface lies **in the surface**, we have a potential inconsistency between the boundary condition and the solution propagated along the characteristic. We are then also unable to extend φ away from the boundary surface because the data on the surface is insufficient to yield values of the derivatives that are needed for a Taylor expansion. To see this, note that the derivatives $\partial\varphi/\partial x$, $\partial\varphi/\partial y$, and $\partial\varphi/\partial z$ can only be determined if we can find two directions (parametrically designated l and l') such that we can solve simultaneously Eq. (9.10) and

$$\frac{\partial\varphi}{\partial l}=\frac{\partial\varphi}{\partial x}\frac{dx}{dl}+\frac{\partial\varphi}{\partial y}\frac{dy}{dl}+\frac{\partial\varphi}{\partial z}\frac{dz}{dl},$$

$$\frac{\partial\varphi}{\partial l'}=\frac{\partial\varphi}{\partial x}\frac{dx}{dl'}+\frac{\partial\varphi}{\partial y}\frac{dy}{dl'}+\frac{\partial\varphi}{\partial z}\frac{dz}{dl'}.$$

A solution can be obtained only if

$$D=\begin{vmatrix} \dfrac{dx}{dl} & \dfrac{dy}{dl} & \dfrac{dz}{dl} \\ \dfrac{dx}{dl'} & \dfrac{dy}{dl'} & \dfrac{dz}{dl'} \\ a & b & c \end{vmatrix}\neq 0.$$

If a characteristic, with $dx/dl''=a$, $dy/dl''=b$, and $dz/dl''=c$, lies in the two-dimensional (2-D) surface, there will only be one further linearly independent direction l, and D will necessarily be zero.

Summarizing, our earlier observations extend to the 3-D case:

A boundary condition is effective in determining a unique solution to a first-order PDE only if the boundary does not include a characteristic, and inconsistencies may arise if a characteristic intersects a boundary more than once.

Exercises

Find the general solutions of the PDEs in Exercises 9.2.1 to 9.2.4.

9.2.1 $\dfrac{\partial\psi}{\partial x}+2\dfrac{\partial\psi}{\partial y}+(2x-y)\psi=0.$

9.2.2 $\dfrac{\partial\psi}{\partial x}-2\dfrac{\partial\psi}{\partial y}+x+y=0.$

9.2.3 $\frac{\partial\psi}{\partial x}+\frac{\partial\psi}{\partial y}=\frac{\partial\psi}{\partial z}$.

9.2.4 $\frac{\partial\psi}{\partial x}+\frac{\partial\psi}{\partial y}+\frac{\partial\psi}{\partial z}=x-y$.

9.2.5 (a) Show that the PDE

$$y\frac{\partial\psi}{\partial x}+x\frac{\partial\psi}{\partial y}=0$$

can be transformed into a readily soluble form by writing it in the new variables $u=xy$, $v=x^2-y^2$, and find its general solution.

(b) Discuss this result in terms of characteristics.

9.2.6 Find the general solution to the PDE

$$x\frac{\partial\psi}{\partial x}-y\frac{\partial\psi}{\partial y}=0.$$

Hint. The solution to Exercise 9.2.5 may provide a suggestion as to how to proceed.

9.3 SECOND-ORDER EQUATIONS

Classes of PDEs

We consider here extending the notion of characteristics to second-order PDEs. This can sometimes be done in a useful fashion. As a preliminary example, consider the following homogeneous second-order equation

$$a^2\frac{\partial^2\varphi(x,y)}{\partial x^2}-c^2\frac{\partial^2\varphi(x,y)}{\partial y^2}=0, \tag{9.11}$$

where a and c are assumed to be real. This equation can be written in the factored form

$$\left[a\frac{\partial}{\partial x}+c\frac{\partial}{\partial y}\right]\left[a\frac{\partial}{\partial x}-c\frac{\partial}{\partial y}\right]\varphi=0, \tag{9.12}$$

and, since the two operator factors commute, we see that Eq. (9.12) will be satisfied if φ is a solution to either of the first-order equations

$$a\frac{\partial\varphi}{\partial x}+c\frac{\partial\varphi}{\partial y}=0 \quad \text{or} \quad a\frac{\partial\varphi}{\partial x}-c\frac{\partial\varphi}{\partial y}=0. \tag{9.13}$$

However, these first-order equations are of just the type discussed in the preceding subsection, so we can identify their respective general solutions as

$$\varphi_1(x,y)=f(cx-ay), \quad \varphi_2(x,y)=g(cx+ay), \tag{9.14}$$

where f and g are arbitrary (and totally unrelated) functions. Moreover, we can identify the stream lines of $ax+cy$ and $ax-cy$ as characteristics, with implications as to the effectiveness and possible consistency of boundary conditions. For some PDEs with second derivatives as given in Eq. (9.11), it will also be practical to propagate solutions along the characteristics.

Look next at the superficially similar equation

$$a^2\frac{\partial^2\varphi(x,y)}{\partial x^2}+c^2\frac{\partial^2\varphi(x,y)}{\partial y^2}=0, \tag{9.15}$$

with a and c again assumed to be real. If we factor this, we get

$$\left[a\frac{\partial}{\partial x}+ic\frac{\partial}{\partial y}\right]\left[a\frac{\partial}{\partial x}-ic\frac{\partial}{\partial y}\right]\varphi=0. \tag{9.16}$$

This factorization is of less practical value, as it leads to **complex** characteristics, which do not have an obvious relevance to boundary conditions. In addition, propagation along such characteristics does not provide a solution to the PDE for physically relevant (i.e., real) coordinate values.

It is customary to identify second-order PDEs as **hyperbolic** if they are of (or can be transformed into) the form given in Eq. (9.11), with real values of a and c. PDEs that are of (or can be transformed into) the form given in Eq. (9.15) are called **elliptic**. The designation is useful because it correlates with the existence (or nonexistence) of real characteristics, and therefore with the behavior of the PDE relative to boundary conditions, with further implications as to convenient methods for solving the PDE. The terms *elliptic* and *hyperbolic* have been introduced based on an analogy to quadratic forms, where $a^2x^2+c^2y^2=d$ is the equation of an ellipse, while $a^2x^2-c^2y^2=d$ is that of a hyperbola.

More general PDEs will have second derivatives of the differential form

$$\mathcal{L}=a\frac{\partial^2\varphi}{\partial x^2}+2b\frac{\partial^2\varphi}{\partial x\partial y}+c\frac{\partial^2\varphi}{\partial y^2}. \tag{9.17}$$

The form in Eq. (9.17) has the following factorization:

$$\mathcal{L}=\left(\frac{b+\sqrt{b^2-ac}}{c^{1/2}}\frac{\partial}{\partial x}+c^{1/2}\frac{\partial}{\partial y}\right)\left(\frac{b-\sqrt{b^2-ac}}{c^{1/2}}\frac{\partial}{\partial x}+c^{1/2}\frac{\partial}{\partial y}\right). \tag{9.18}$$

Equation (9.18) is easily verified by expanding the product. The equation also shows that the characteristics of Eq. (9.17) are real if and only if $b^2-ac\geq 0$. This quantity is well known from elementary algebra, being the **discriminant** of the quadratic form $at^2+2bt+c$. If $b^2-ac>0$, the two factors identify two linearly independent real characteristics, as were found for the prototype hyperbolic PDE discussed in Eqs. (9.11) to (9.14). If $b^2-ac<0$, the characteristics will, as for the prototype elliptic PDE in Eqs. (9.15) and (9.16), form a complex conjugate pair. We now have, however, one new possibility: If $b^2-ac=0$ (a case that for quadratic forms is that of a parabola), we have a PDE that has exactly one linearly independent characteristic; such PDEs are termed **parabolic**, and the canonical form adopted for them is

$$a\frac{\partial\varphi}{\partial x}=\frac{\partial^2\varphi}{\partial y^2}. \tag{9.19}$$

If the original PDE lacked a $\partial/\partial x$ term, it would in effect be an ODE in y that depends on x only parametrically and need not be considered further in the context of methods for PDEs.

To complete our discussion of the second-order form in Eq. (9.17), we need to show that it can be transformed into the canonical form for the PDE of its classification. For this purpose we consider the transformation to new variables ξ, η, defined as

$$\xi = c^{1/2}x - c^{-1/2}by, \quad \eta = c^{-1/2}y. \tag{9.20}$$

By systematic application of the chain rule to evaluate $\partial^2/\partial x^2$, $\partial^2/\partial x\partial y$, and $\partial^2/\partial y^2$, it can be shown that

$$\mathcal{L} = (ac - b^2)\frac{\partial^2\varphi}{\partial\xi^2} + \frac{\partial^2\varphi}{\partial\eta^2}. \tag{9.21}$$

Verification of Eq. (9.21) is the subject of Exercise 9.3.1.

Equation (9.21) shows that the classification of our PDE remains invariant under transformation, and is hyperbolic if $b^2 - ac > 0$, elliptic if $b^2 - ac < 0$, and parabolic if $b^2 - ac = 0$. Perhaps better seen from Eq. (9.18), we see that the stream lines of the characteristics have slope

$$\frac{dy}{dx} = \frac{c}{b \pm \sqrt{b^2 - ac}}. \tag{9.22}$$

More than Two Independent Variables

While we will not carry out a full analysis, it is important to note that many problems in physics involve more than two dimensions (often, three spatial dimensions or several spatial dimensions plus time). Often, the behavior in the multiple spatial dimensions is similar, and we apply the terms *hyperbolic*, *elliptic*, and *parabolic* in a way that relates the spatial to the time derivatives when the latter occur. Thus, these equations are classified as indicated:

Laplace equation	$\nabla^2\psi = 0$	elliptic
Poisson equation	$\nabla^2\psi = -\rho/\varepsilon_0$	elliptic
Wave equation	$\nabla^2\psi = \dfrac{1}{c^2}\dfrac{\partial^2\psi}{\partial t^2}$	hyperbolic
Diffusion equation	$a\dfrac{\partial\psi}{\partial t} = \nabla^2\psi$	parabolic

The specific equations mentioned here are very important in physics and will be further discussed in later sections of this chapter. These examples, of course, do not represent the full range of second-order PDEs, and do not include cases where the coefficients in the differential operator are functions of the coordinates. In that case, the classification into elliptic, hyperbolic, and parabolic is only local; the class may change as the coordinates vary.

Boundary Conditions

Usually, when we know a physical system at some time and the law governing the physical process, then we are able to predict the subsequent development. Such initial values are the most common boundary conditions associated with ODEs and PDEs. Finding solutions that match given points, curves, or surfaces corresponds to boundary value problems. Solutions usually are required to satisfy certain imposed (for example, asymptotic) boundary conditions. These boundary conditions ordinarily take one of three forms:

1. **Cauchy boundary conditions**. The value of a function and normal derivative specified on the boundary. In electrostatics this would mean φ, the potential, and E_n, the normal component of the electric field.
2. **Dirichlet boundary conditions**. The value of a function specified on the boundary. In electrostatics, this would mean the potential φ.
3. **Neumann boundary conditions**. The normal derivative (normal gradient) of a function specified on the boundary. In the electrostatic case this would be E_n and therefore σ, the surface charge density.

Because the three classes of second-order PDEs have different patterns of characteristics, the boundary conditions needed to specify (in a consistent way) a unique solution will depend on the equation class. An exact analysis of the role of boundary conditions is complicated and beyond the scope of the present text. However, a summary of the relation of these three types of boundary conditions to the three classes of 2-D partial differential equations is given in Table 9.1. For a more extended discussion of these partial differential equations the reader may consult Morse and Feshbach, Chapter 6 (see Additional Readings).

Parts of Table 9.1 are simply a matter of maintaining internal consistency or of common sense. For instance, for Poisson's equation with a closed surface, Dirichlet conditions lead

Table 9.1 Relation between PDE and Boundary Conditions

Boundary Conditions	Class of Partial Differential Equation		
	Elliptic	Hyperbolic	Parabolic
	Laplace, Poisson in (x, y)	Wave equation in (x, t)	Diffusion equation in (x, t)
Cauchy			
Open surface	Unphysical results (instability)	**Unique, stable solution**	Too restrictive
Closed surface	Too restrictive	Too restrictive	Too restrictive
Dirichlet			
Open surface	Insufficient	Insufficient	**Unique, stable solution** in one direction
Closed surface	**Unique, stable solution**	Solution not unique	Too restrictive
Neumann			
Open surface	Insufficient	Insufficient	**Unique, stable solution** in one direction
Closed surface	**Unique, stable solution**	Solution not unique	Too restrictive

to a unique, stable solution. Neumann conditions, independent of the Dirichlet conditions, likewise lead to a unique stable solution independent of the Dirichlet solution. Therefore, Cauchy boundary conditions (meaning Dirichlet plus Neumann) could lead to an inconsistency.

The term **boundary conditions** includes as a special case the concept of **initial conditions**. For instance, specifying the initial position x_0 and the initial velocity v_0 in some dynamical problem would correspond to the Cauchy boundary conditions. Note, however, that an initial condition corresponds to applying the condition at only one end of the allowed range of the (time) variable.

Finally, we note that Table 9.1 oversimplifies the situation in various ways. For example, the Helmholtz PDE,

$$\nabla^2\psi \pm k^2\psi = 0,$$

(which could be thought of as the reduction of a parabolic time-dependent equation to its spatial part) has solution(s) for Dirichlet conditions on a closed boundary only for certain values of its parameter k. The determination of k and the characterization of these solutions is an eigenvalue problem and is important for physics.

Nonlinear PDEs

Nonlinear ODEs and PDEs are a rapidly growing and important field. We encountered earlier the simplest linear wave equation,

$$\frac{\partial\psi}{\partial t} + c\frac{\partial\psi}{\partial x} = 0,$$

as the first-order PDE of the wavefronts of the wave equation. The simplest nonlinear wave equation,

$$\frac{\partial\psi}{\partial t} + c(\psi)\frac{\partial\psi}{\partial x} = 0, \tag{9.23}$$

results if the local speed of propagation, c, is not constant but depends on the wave ψ. When a nonlinear equation has a solution of the form $\psi(x,t) = A\cos(kx - \omega t)$, where $\omega(k)$ varies with k so that $\omega''(k) \neq 0$, then it is called **dispersive**. Perhaps the best-known nonlinear dispersive equation is the **Korteweg-deVries** equation,

$$\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi}{\partial x} + \frac{\partial^3\psi}{\partial x^3} = 0, \tag{9.24}$$

which models the lossless propagation of shallow water waves and other phenomena. It is widely known for its **soliton** solutions. A soliton is a traveling wave with the property of persisting through an interaction with another soliton: After they pass through each other, they emerge in the same shape and with the same velocity and acquire no more than a phase shift. Let $\psi(\xi = x - ct)$ be such a traveling wave. When substituted into Eq. (9.24) this yields the nonlinear ODE

$$(\psi - c)\frac{d\psi}{d\xi} + \frac{d^3\psi}{d\xi^3} = 0, \tag{9.25}$$

which can be integrated to yield

$$\frac{d^2\psi}{d\xi^2} = c\psi - \frac{\psi^2}{2}. \tag{9.26}$$

There is no additive integration constant in Eq. (9.26), because the solution must be such that $d^2\psi/d\xi^2 \to 0$ with $\psi \to 0$ for large ξ. This causes ψ to be localized at the characteristic $\xi = 0$, or $x = ct$. Multiplying Eq. (9.26) by $d\psi/d\xi$ and integrating again yields

$$\left(\frac{d\psi}{d\xi}\right)^2 = c\psi^2 - \frac{\psi^3}{3}, \tag{9.27}$$

where $d\psi/d\xi \to 0$ for large ξ. Taking the root of Eq. (9.27) and integrating again yields the soliton solution

$$\psi(x - ct) = \frac{3c}{\cosh^2\left(\frac{1}{2}\sqrt{c}(x - ct)\right)}. \tag{9.28}$$

Exercises

9.3.1 Show that by making a change of variables to $\xi = c^{1/2}x - c^{-1/2}by$, $\eta = c^{-1/2}y$, the operator $\mathcal{L}$ of Eq. (9.18) can be brought to the form

$$\mathcal{L} = (ac - b^2)\frac{\partial^2}{\partial\xi^2} + \frac{\partial^2}{\partial\eta^2}.$$

9.4 Separation of Variables

Partial differential equations are clearly important in physics, as evidenced by the PDEs listed in Section 9.1, and of equal importance is the development of methods for their solution. Our discussion of characteristics has suggested an approach that will be useful for some problems. Other general techniques for solving PDEs can be found, for example, in the books by Bateman and by Gustafson listed in the Additional Readings at the end of this chapter. However, the technique described in the present section is probably that most widely used.

The method developed in this section for solution of a PDE splits a partial differential equation of n variables into n ordinary differential equations, with the intent that an overall solution to the PDE will be a product of single-variable functions which are solutions to the individual ODEs. In problems amenable to this method, the boundary conditions are usually such that they separate at least partially into conditions that can be applied to the separate ODEs.

Further discussion of the method depends on the nature of the problem we seek to solve, so we now make the observation that PDEs occur in physics in two contexts, either as

- An equation with no unknown parameters for which there is expected to be a unique solution consistent with the boundary conditions (typical example: Laplace equation for the electrostatic potential with the potential specified on the boundary), or

- An eigenvalue problem which will have solutions consistent with the boundary conditions only for certain values of an embedded but initially unknown parameter (the eigenvalue).

In the first of these two cases, the unique solution is typically approached by first applying boundary conditions to the separate ODEs to specialize their solutions as much as possible. The solution is at this point normally not unique, and we have a (usually infinite) number of product solutions that satisfy the boundary conditions thus far applied. We then regard these product solutions as a basis that can be used to form an expansion that satisfies the remaining boundary condition(s). We illustrate with the first and fourth examples of this section.

In the second case identified above, we typically have homogeneous boundary conditions (solution equal to zero on the boundary), and in favorable situations can satisfy all the boundary conditions by imposing them on the separate ODEs. At this point we usually find that each product solves our PDE with a different value of its embedded parameter, so that we are obtaining eigenfunctions and eigenvalues. This process is illustrated in the second and third examples of the present section.

The method of separation of variables proceeds by dividing the PDE into pieces each of which can be set equal to a **constant of separation**. If our PDE has n independent variables, there will be $n-1$ independent separation constants (though we often prefer a more symmetric formulation with n separation constants plus an equation connecting them). The separation constants may have values that are restricted by invoking boundary conditions.

To get a broad understanding of the method of separation of variables, it is useful to see how it is carried out in a variety of coordinate systems. Here we examine the process in Cartesian, cylindrical, and spherical polar coordinates. For application to other coordinate systems we refer the reader to the second edition of this text.

Cartesian Coordinates

In Cartesian coordinates the Helmholtz equation becomes

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+k^2\psi=0, \tag{9.29}$$

using Eq. (3.62) for the Laplacian. For the present, let k^2 be a constant. As stated in the introductory paragraphs of this section, our strategy will be to split Eq. (9.29) into a set of ordinary differential equations. To do so, let

$$\psi(x,y,z)=X(x)Y(y)Z(z) \tag{9.30}$$

and substitute back into Eq. (9.29). How do we know Eq. (9.30) is valid? When the differential operators in various variables are additive in the PDE, that is, when there are no products of differential operators in different variables, the separation method has a chance to succeed. For success, it is usually also necessary that at least some of the boundary conditions separate into conditions on the separate factors. At any rate, we are proceeding in the spirit of let's try and see if it works. If our attempt succeeds, then Eq. (9.30) will be justified. If it does not succeed, we shall find out soon enough and then we can try another attack, such as Green's functions, integral transforms, or brute-force numerical analysis. With ψ assumed given by Eq. (9.30), Eq. (9.29) becomes

$$YZ\frac{d^2X}{dx^2} + XZ\frac{d^2Y}{dy^2} + XY\frac{d^2Z}{dz^2} + k^2XYZ = 0. \tag{9.31}$$

Dividing by $\psi = XYZ$ and rearranging terms, we obtain

$$\frac{1}{X}\frac{d^2X}{dx^2} = -k^2 - \frac{1}{Y}\frac{d^2Y}{dy^2} - \frac{1}{Z}\frac{d^2Z}{dz^2}. \tag{9.32}$$

Equation (9.32) exhibits one separation of variables. The left-hand side is a function of x alone, whereas the right-hand side depends only on y and z and not on x. But x, y, and z are all independent coordinates. The equality of two sides that depend on different variables can only be attained if each side must be equal to the same constant, a constant of separation. We choose[2]

$$\frac{1}{X}\frac{d^2X}{dx^2} = -l^2, \tag{9.33}$$

$$-k^2 - \frac{1}{Y}\frac{d^2Y}{dy^2} - \frac{1}{Z}\frac{d^2Z}{dz^2} = -l^2. \tag{9.34}$$

Now, turning our attention to Eq. (9.34), we obtain

$$\frac{1}{Y}\frac{d^2Y}{dy^2} = -k^2 + l^2 - \frac{1}{Z}\frac{d^2Z}{dz^2}, \tag{9.35}$$

and a second separation has been achieved. Here we have a function of y equated to a function of z. We resolve it, as before, by equating each side to another constant of separation, $-m^2$,

$$\frac{1}{Y}\frac{d^2Y}{dy^2} = -m^2, \tag{9.36}$$

$$-k^2 + l^2 - \frac{1}{Z}\frac{d^2Z}{dz^2} = -m^2. \tag{9.37}$$

The separation is now complete, but to make the formulation more symmetrical, we will set

$$\frac{1}{Z}\frac{d^2Z}{dz^2} = -n^2, \tag{9.38}$$

and then consistency with Eq. (9.37) leads to the condition

$$l^2 + m^2 + n^2 = k^2. \tag{9.39}$$

Now we have three ODEs, Eqs. (9.33), (9.36), and (9.38), to replace Eq. (9.29). Our assumption, Eq. (9.30), has succeeded in splitting the PDE; if we can also use the factored form to satisfy the boundary conditions, our solution of the PDE will be complete.

[2]The choice of sign for separation constants is completely arbitrary, and will be fixed in specific problems by the need to satisfy specific boundary conditions, and particularly to avoid the unnecessary introduction of complex numbers.

It is convenient to label the solution according to the choice of our constants l, m, and n; that is,

$$\psi_{lmn}(x, y, z) = X_l(x)Y_m(y)Z_n(z). \tag{9.40}$$

Subject to the boundary conditions of the problem being solved and to the condition $k^2 = l^2 + m^2 + n^2$, we may choose l, m, and n as we like, and Eq. (9.40) will still be a solution of Eq. (9.29), provided only that $X_l(x)$ is a solution of Eq. (9.33), and so on. Because our original PDE is homogeneous and linear, we may develop **the most general solution** of Eq. (9.29) by taking a **linear combination of solutions** ψ_{lmn},

$$\Psi = \sum_{l,m} a_{lm}\psi_{lmn}, \tag{9.41}$$

where it is understood that n will be given a value consistent with Eq. (9.39) and with the values of l and m.

Finally, the constant coefficients a_{lm} must be chosen to permit Ψ to satisfy the boundary conditions of the problem, leading usually to a discrete set of values l, m.

Reviewing what we have done, it can be seen that the separation into ODEs could still have been achieved if k^2 were replaced by any function that depended additively on the variables, i.e., if

$$k^2 \longrightarrow f(x) + g(y) + h(z).$$

A case of practical importance would be the choice $k^2 \longrightarrow C(x^2 + y^2 + z^2)$, leading to the problem of a 3-D quantum harmonic oscillator. Replacing the constant term k^2 by a separable function of the variables will, of course, change the ODEs we obtain in the separation process and may have implications relative to the boundary conditions.

Example 9.4.1 Laplace Equation for a Parallelepiped

As a concrete example we take Eq. (9.29) with $k = 0$, which makes it a Laplace equation, and ask for its solution in a parallelepiped defined by the planar surfaces $x = 0$, $x = c$, $y = 0$, $y = c$, $z = 0$, $z = L$, with the Dirichlet boundary condition $\psi = 0$ on all the boundaries except that at $z = L$; on that boundary ψ is given the constant value V. See Fig. 9.1. This is a problem in which the PDE contains no unknown parameters and should have a unique solution.

We expect a solution of the generic form given by Eq. (9.41), with ψ_{lmn} given by Eq. (9.40). To proceed further, we need to develop the actual functional forms of $X(x)$, $Y(y)$, and $Z(z)$. For X and Y, the ODEs, written in conventional form, are

$$X'' = -l^2 X, \qquad Y'' = -m^2 Y,$$

with general solutions

$$X = A\sin lx + B\cos lx, \qquad Y = A'\sin my + B'\cos my.$$

We could have written X and Y as complex exponentials, but that choice would be less convenient when we consider the boundary conditions. To satisfy the boundary condition at $x = 0$, we set $X(0) = 0$, which can be accomplished by choosing $B = 0$; to satisfy

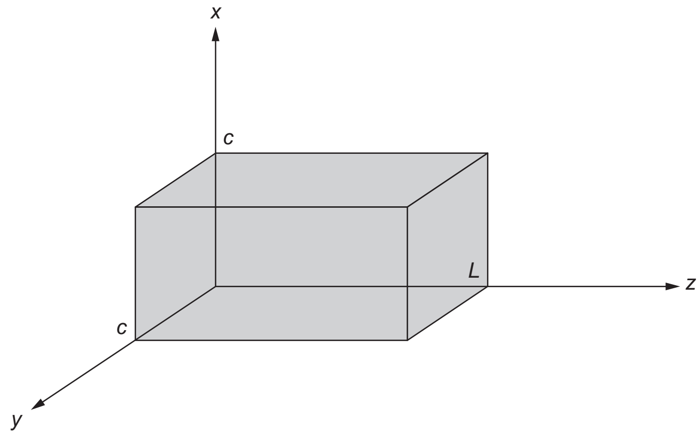

FIGURE 9.1 Parallelepiped for solution of Laplace equation.

the boundary condition at $x = c$, we set $X(c) = 0$, which causes us to choose l such that $lc = \lambda\pi$, where λ must be a nonzero integer. Without loss of generality, we can restrict λ to positive values, as $-X$ and X are linearly dependent. Moreover, we can include whatever scale factor is ultimately needed in our solution for $Z(z)$, so we may set $A = 1$. Similar remarks apply to the solution $Y(y)$, so our solutions for X and Y take the final form

$$X_\lambda(x) = \sin\left(\frac{\lambda\pi x}{c}\right), \quad Y_\mu(y) = \sin\left(\frac{\mu\pi y}{c}\right), \tag{9.42}$$

with $\lambda = 1, 2, 3, \ldots$ and $\mu = 1, 2, 3, \ldots$.

Next we consider the ODE for Z. It must be solved with a value of n^2, calculated from Eq. (9.39) with $k = 0$ as

$$n^2 = -\frac{\pi^2}{c^2}(\lambda^2 + \mu^2).$$

This equation suggests that n will be imaginary, but that is unimportant here. Returning to the ODE for Z, we now see that it becomes

$$Z'' = +\frac{\pi^2}{c^2}(\lambda^2 + \mu^2)Z,$$

and the general solution for $Z(z)$ for given λ and μ is then easily identified as

$$Z_{\lambda\mu}(z) = A\, e^{\rho_{\lambda\mu} z} + B\, e^{-\rho_{\lambda\mu} z}, \quad \text{with } \rho_{\lambda\mu} = \frac{\pi}{c}\sqrt{\lambda^2 + \mu^2}. \tag{9.43}$$

We now specialize Eq. (9.43) in a way that makes $Z_{\lambda\mu}(0) = 0$ and $Z_{\lambda\mu}(L) = V$. Noting that $\sinh(\rho_{\lambda\mu} z)$ is a linear combination of $e^{\rho_{\lambda\mu} z}$ and $e^{-\rho_{\lambda\mu} z}$, we write

$$Z_{\lambda\mu}(z) = V\frac{\sinh(\rho_{\lambda\mu} z)}{\sinh(\rho_{\lambda\mu} L)}. \tag{9.44}$$

At this point, we have made choices that cause all the boundary conditions to be satisfied except that at $z = L$, and we are now ready to select the coefficients $a_{\lambda\mu}$ as required by the remaining boundary condition, which because of Eq. (9.44) corresponds to

$$\frac{1}{V}\Psi(x,y,L)=\sum_{\lambda\mu}a_{\lambda\mu}\sin\left(\frac{\lambda\pi x}{c}\right)\sin\left(\frac{\mu\pi y}{c}\right)=1. \tag{9.45}$$

The symmetry of this expression suggests that we write $a_{\lambda\mu}=b_\lambda b_\mu$, and find the coefficients b_λ from the equation

$$\sum_\lambda b_\lambda \sin\left(\frac{\lambda\pi x}{c}\right)=1. \tag{9.46}$$

Because the sine functions in Eq. (9.46) are the eigenfunctions of the one-dimensional (1-D) equation for X, which is a Hermitian eigenproblem, they form an orthogonal set on the interval $(0,c)$, so the b_λ can be computed by the following formulas:

$$\begin{aligned}
b_\lambda &= \frac{\left\langle \sin\left(\frac{\lambda\pi x}{c}\right)\Big| 1\right\rangle}{\left\langle \sin\left(\frac{\lambda\pi x}{c}\right)\Big| \sin\left(\frac{\lambda\pi x}{c}\right)\right\rangle}=\frac{\int\limits_0^c \sin(\lambda\pi x/c)\,dx}{\int\limits_0^c \sin^2(\lambda\pi x/c)\,dx}\\
&=\frac{4}{\lambda\pi},\qquad \lambda \text{ odd},\\
&=0,\qquad \lambda \text{ even},
\end{aligned}$$

and our complete solution for the potential in the parallelepiped becomes

$$\Psi(x,y,z)=V\sum_{\lambda\mu}b_\lambda b_\mu \sin\left(\frac{\lambda\pi x}{c}\right)\sin\left(\frac{\mu\pi y}{c}\right)\frac{\sinh(\rho_{\lambda\mu}z)}{\sinh(\rho_{\lambda\mu}L)}. \tag{9.47}$$

■

As briefly mentioned earlier, PDEs also occur as eigenvalue problems. Here is a simple example.

Example 9.4.2 Quantum Particle in a Box

We consider a particle of mass m trapped in a box with planar faces at $x=0$, $x=a$, $y=0$, $y=b$, $z=0$, $z=c$. The quantum stationary states of this system are the eigenfunctions of the Schrödinger equation

$$-\frac{1}{2}\nabla^2\psi(x,y,z)=E\psi(x,y,z), \tag{9.48}$$

where this PDE is subject to the Dirichlet boundary condition $\psi=0$ on the walls of the box. We identify E as the stationary-state energy (the eigenvalue), in a system of units with $m=\hbar=1$. This is a Helmholtz equation with the new wrinkle that E is not initially known. The boundary conditions are such that this PDE has no solution except for a set of discrete values of E. We want to find both those values and the corresponding eigenfunctions.

Separating the variables in Eq. (9.48) by assuming a solution of the form Eq. (9.30), the PDE becomes

$$-\left(\frac{X''}{X}+\frac{Y''}{Y}+\frac{Z''}{Z}\right)=2E, \tag{9.49}$$

and the separation yields

$$\frac{X''}{X}=-l^2, \quad \text{with solution } \ X=A\,\sin lx+B\cos lx.$$

After applying the boundary conditions at $x=0$ and $x=a$ we get (scaling to $A=1$)

$$X_\lambda=\sin\left(\frac{\lambda\pi x}{a}\right), \quad \lambda=1,2,3,\ldots, \quad \text{so } \ l=\lambda\pi/a. \tag{9.50}$$

Because the X equation is a 1-D Hermitian eigenvalue problem, these functions $X_\lambda(x)$ are orthogonal on $0\le x\le a$.

Similar processing of the Y and Z equations, with separation constants $-m^2$ and $-n^2$, yields

$$\begin{aligned} Y_\mu&=\sin\left(\frac{\mu\pi y}{b}\right), \quad \mu=1,2,3,\ldots, \quad \text{so } \ m=\mu\pi/b,\\ Z_\nu&=\sin\left(\frac{\nu\pi z}{c}\right), \quad \nu=1,2,3,\ldots, \quad \text{so } \ n=\nu\pi/c, \end{aligned} \tag{9.51}$$

yielding two additional 1-D eigenvalue problems.

Replacing X''/X, Y''/Y, Z''/Z in Eq. (9.49), respectively, by $-l^2$, $-m^2$, $-n^2$, and then evaluating these quantities from Eqs. (9.50) and (9.51), we have

$$l^2+m^2+n^2=2E, \quad \text{or} \quad E=\frac{\pi^2}{2}\left(\frac{\lambda^2}{a^2}+\frac{\mu^2}{b^2}+\frac{\nu^2}{c^2}\right), \tag{9.52}$$

with λ, μ, and μ arbitrary positive integers. The situation is quite different from our solution, Example 9.4.1, of the Laplace equation. Instead of a unique solution we have an infinite set of solutions, corresponding to all positive integer triples (λ,μ,ν), each with its own value of E. Making the observation that the differential operator on the left-hand side of Eq. (9.47) is Hermitian in the presence of the chosen boundary conditions, we have found a complete orthogonal set of its eigenfunctions. The orthogonality is obvious, as it can be confirmed from the orthogonality of the X_λ, Y_μ, and Z_ν on their respective 1-D intervals. Because we set the coefficients of all the sine functions to unity, our overall eigenfunctions are not normalized, but we can easily normalize them if we so choose.

We close this example with the observation that this boundary-value problem will not have a solution for arbitrarily chosen values of E, as the E values must satisfy Eq. (9.52) with **integer values** of λ, μ, and ν. This will cause the E values of the problem solutions to be a discrete set; using terminology introduced in a previous chapter, our boundary-value problem can be said to have a **discrete spectrum**. ■

Circular Cylindrical Coordinates

Curvilinear coordinate systems introduce additional nuances into the process for separating variables. Again we consider the Helmholtz equation, now in circular cylindrical coordinates. With our unknown function ψ dependent on ρ, φ, and z, that equation becomes, using Eq. (3.149) for ∇^2:

$$\nabla^2\psi(\rho,\varphi,z) + k^2\psi(\rho,\varphi,z) = 0, \tag{9.53}$$

or

$$\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right) + \frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\varphi^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0. \tag{9.54}$$

As before, we assume a factored form[3] for ψ,

$$\psi(\rho,\varphi,z) = P(\rho)\Phi(\varphi)Z(z). \tag{9.55}$$

Substituting into Eq. (9.46), we have

$$\frac{\Phi Z}{\rho}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + \frac{PZ}{\rho^2}\frac{d^2\Phi}{d\varphi^2} + P\Phi\frac{d^2Z}{dz^2} + k^2P\Phi Z = 0. \tag{9.56}$$

All the partial derivatives have become ordinary derivatives. Dividing by $P\Phi Z$ and moving the z derivative to the right-hand side yields

$$\frac{1}{\rho P}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + \frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2} + k^2 = -\frac{1}{Z}\frac{d^2Z}{dz^2}. \tag{9.57}$$

Again, a function of z on the right appears to depend on a function of ρ and φ on the left. We resolve this by setting each side of Eq. (9.57) equal to the same constant. Let us choose[4] $-l^2$. Then

$$\frac{d^2Z}{dz^2} = l^2Z \tag{9.58}$$

and

$$\frac{1}{\rho P}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + \frac{1}{\rho^2\Phi}\frac{d^2\Phi}{d\varphi^2} + k^2 = -l^2. \tag{9.59}$$

Setting

$$k^2 + l^2 = n^2, \tag{9.60}$$

multiplying by ρ^2, and rearranging terms, we obtain

$$\frac{\rho}{P}\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + n^2\rho^2 = -\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2}. \tag{9.61}$$

[3]For those with limited familiarity with the Greek alphabet, we point out that the symbol P is the upper-case form of ρ.

[4]Again, the choice of sign of the separation constant is arbitrary. However, the minus sign chosen for the axial coordinate z is optimum if we expect exponential dependence on z, from Eq. (9.58). A positive sign is chosen for the azimuthal coordinate φ in expectation of a periodic dependence on φ, from Eq. (9.62).

We set the right-hand side equal to m^2, so

$$\frac{d^2\Phi}{d\varphi^2} = -m^2\Phi, \tag{9.62}$$

and the left-hand side of Eq. (9.61) rearranges into a separate equation for ρ:

$$\rho\frac{d}{d\rho}\left(\rho\frac{dP}{d\rho}\right) + (n^2\rho^2 - m^2)P = 0. \tag{9.63}$$

Typically, Eq. (9.62) will be subject to the boundary condition that Φ have periodicity 2π and will therefore have solutions

$$e^{\pm im\varphi} \text{ or, equivalently } \sin m\varphi,\ \cos m\varphi, \text{ with integer } m.$$

The ρ equation, Eq. (9.63), is Bessel's differential equation (in the independent variable $n\rho$), originally encountered in Chapter 7. Because of its occurrence here (and in many other places relevant to physics), it warrants extensive study and is the topic of Chapter 14. The separation of variables of Laplace's equation in parabolic coordinates also gives rise to Bessel's equation. It may be noted that the Bessel equation is notorious for the variety of disguises it may assume. For an extensive tabulation of possible forms the reader is referred to *Tables of Functions* by Jahnke and Emde.[5]

Summarizing, we have found that the original Helmholtz equation, a 3-D PDE, can be replaced by three ODEs, Eqs. (9.58), (9.62), and (9.63). Noting that the ODE for ρ contains the separation constants from the z and φ equations, the solutions we have obtained for the Helmholtz equation can be written, with labels, as

$$\psi_{lm}(\rho,\varphi,z) = P_{lm}(\rho)\Phi_m(\varphi)Z_l(z), \tag{9.64}$$

where we probably should recall that the n in Eq. (9.63) for P is a function of l (specifically, $n^2 = l^2 + k^2$). The most general solution of the Helmholtz equation can now be constructed as a linear combination of the product solutions:

$$\Psi(\rho,\varphi,z) = \sum_{l,m} a_{lm}P_{lm}(\rho)\Phi_m(\varphi)Z_l(z). \tag{9.65}$$

Reviewing what we have done, we note that the separation could still have been achieved if k^2 had been replaced by any additive function of the form

$$k^2 \quad\longrightarrow\quad f(r) + \frac{g(\varphi)}{\rho^2} + h(z).$$

Example 9.4.3 Cylindrical Eigenvalue Problem

In this example we regard Eq. (9.53) as an eigenvalue problem, with Dirichlet boundary conditions $\psi = 0$ on all boundaries of a finite cylinder, with k^2 initially unknown and to be determined. Our region of interest will be a cylinder with curved boundaries at $\rho = R$ and with end caps at $z = \pm L/2$, as shown in Fig. 9.2. To emphasize that k^2 is an eigenvalue,

[5] E. Jahnke and F. Emde, *Tables of Functions*, 4th rev. ed., New York: Dover (1945), p. 146; also, E. Jahnke, F. Emde, and F. Lösch, *Tables of Higher Functions*, 6th ed., New York: McGraw-Hill (1960).

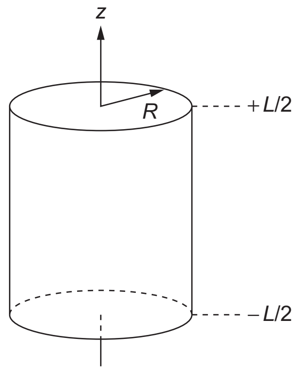

FIGURE 9.2 Cylindrical region for solution of the Helmholtz equation.

we rename it λ, and our eigenvalue equation is, symbolically,

$$-\nabla^2\psi = \lambda\psi, \tag{9.66}$$

with boundary conditions $\psi = 0$ at $\rho = R$ and at $z = \pm L/2$. Apart from constants, this is the time-independent Schrödinger equation for a particle in a cylindrical cavity. We limit the present example to the determination of the smallest eigenvalue (the **ground state**). This will be the solution to the PDE with the smallest number of oscillations, so we seek a solution without zeros (**nodes**) in the interior of the cylindrical region.

Again, we seek separated solutions of the form given in Eq. (9.55). The ODEs for Z and Φ, Eqs. (9.58) and (9.62), have the simple forms

$$Z'' = l^2 Z, \quad \Phi'' = -m^2\Phi,$$

with general solutions

$$Z = A\,e^{lz} + B\,e^{-lz}, \quad \Phi = A'\sin m\varphi + B'\cos m\varphi.$$

We now need to specialize these solutions to satisfy the boundary conditions. The condition on Φ is simply that it be periodic in φ with period 2π; this result will be obtained if m is any integer (including $m = 0$, which corresponds to the simple solution $\Phi =$ constant). Since our objective here is to obtain the least oscillatory solution, we choose that form, $\Phi =$ constant, for Φ.

Looking next at Z, we note that the arbitrary choice of sign for the separation constant l^2 has led to a form of solution that appears not to be optimum for fulfilling conditions requiring $Z = 0$ at the boundaries. But, writing $l^2 = -\omega^2$, $l = i\omega$, Z becomes a linear combination of $\sin\omega z$ and $\cos\omega z$; the least oscillatory solution with $Z(\pm L/2) = 0$ is $Z = \cos(\pi z/L)$, so $\omega = \pi/L$, and $l^2 = -\pi^2/L^2$.

The functions $Z(z)$ and $\Phi(\varphi)$ that we have found satisfy the boundary conditions in z and φ but it remains to choose $P(\rho)$ in a way that produces $P = 0$ at $\rho = R$ with the least oscillation in P. The equation governing P, Eq. (9.63), is

$$\rho^2 P'' + \rho P' + n^2\rho^2 P = 0, \tag{9.67}$$

where n was introduced as satisfying (in the current notation) $n^2 = \lambda + l^2$, see Eq. (9.60). Continuing now with Eq. (9.67), we identify as the Bessel equation of order zero in $x = n\rho$. As we learned in Chapter 7, this ODE has two linearly independent solutions, of which only the one designated J_0 is nonsingular at the origin. Since we need here a solution that is regular over the entire range $0 \le x \le nR$, the solution we must choose is $J_0(n\rho)$.

We can now see what is necessary to satisfy the boundary condition at $\rho = R$, namely that $J_0(nR)$ vanish. This is a condition on the parameter n. Remembering that we want the least oscillatory function P, we need for n to be such that nR will be the location of the smallest zero of J_0. Giving this point the name α (which by numerical methods can be found to be approximately 2.4048), our boundary condition takes the form $nR = \alpha$, or $n = \alpha/R$, and our complete solution to the Helmholtz equation can be written

$$\psi(\rho, \varphi, z) = J_0\left(\frac{\alpha\rho}{R}\right)\cos\left(\frac{\pi z}{L}\right). \tag{9.68}$$

To complete our analysis, we must figure out how to arrange that $n = \alpha/R$. Since the condition connecting n, l, and λ rearranges to

$$\lambda = n^2 - l^2, \tag{9.69}$$

we see that the condition on n translates into one on λ. Our PDE has a unique ground-state solution consistent with the boundary conditions, namely an *eigenfunction* whose *eigenvalue* can be computed from Eq. (9.69), yielding

$$\lambda = \frac{\alpha^2}{R^2} + \frac{\pi^2}{L^2}.$$

If we had not restricted consideration to the ground state (by choosing the least oscillatory solution), we would have (in principle) been able to obtain a complete set of eigenfunctions, each with its own eigenvalue. ■

Spherical Polar Coordinates

As a final exercise in the separation of variables in PDEs, let us try to separate the Helmholtz equation, again with k^2 constant, in spherical polar coordinates. Using Eq. (3.158), our PDE is

$$\frac{1}{r^2\sin\theta}\left[\sin\theta\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right] = -k^2\psi. \tag{9.70}$$

Now, in analogy with Eq. (9.30) we try

$$\psi(r, \theta, \varphi) = R(r)\Theta(\theta)\Phi(\varphi). \tag{9.71}$$

By substituting back into Eq. (9.70) and dividing by $R\Theta\Phi$, we have

$$\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{\Theta r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{\Phi r^2\sin^2\theta}\frac{d^2\Phi}{d\varphi^2} = -k^2. \tag{9.72}$$

Note that all derivatives are now ordinary derivatives rather than partials. By multiplying by $r^2 \sin^2\theta$, we can isolate $(1/\Phi)(d^2\Phi/d\varphi^2)$ to obtain

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\varphi^2} = r^2\sin^2\theta\left[-k^2 - \frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{1}{\Theta r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)\right]. \tag{9.73}$$

Equation (9.73) relates a function of φ alone to a function of r and θ alone. Since r, θ, and φ are independent variables, we equate each side of Eq. (9.73) to a constant. In almost all physical problems, φ will appear as an azimuth angle. This suggests a periodic solution rather than an exponential. With this in mind, let us use $-m^2$ as the separation constant, which then must be an integer squared. Then

$$\frac{1}{\Phi}\frac{d^2\Phi(\varphi)}{d\varphi^2} = -m^2 \tag{9.74}$$

and

$$\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{\Theta r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \frac{m^2}{r^2\sin^2\theta} = -k^2. \tag{9.75}$$

Multiplying Eq. (9.75) by r^2 and rearranging terms, we obtain

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + r^2k^2 = -\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{m^2}{\sin^2\theta}. \tag{9.76}$$

Again, the variables are separated. We equate each side to a constant, λ, and finally obtain

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \frac{m^2}{\sin^2\theta}\Theta + \lambda\Theta = 0, \tag{9.77}$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + k^2R - \frac{\lambda R}{r^2} = 0. \tag{9.78}$$

Once more we have replaced a partial differential equation of three variables by three ODEs.

The ODE for Φ is the same as that encountered in cylindrical coordinates, with solutions $\exp(\pm im\varphi)$ or $\sin m\varphi$, $\cos m\varphi$. The Θ ODE can be made less forbidding by changing the independent variable from θ to $t = \cos\theta$, after which Eq. (9.77), with $\Theta(\theta)$ now written as $P(\cos\theta) = P(t)$, becomes

$$(1-t^2)P''(t) - 2tP'(t) - \frac{m^2}{1-t^2}P(t) + \lambda P(t) = 0. \tag{9.79}$$

This is the **associated Legendre equation** (called the **Legendre equation** if $m = 0$), and is discussed in detail in Chapter 15. We normally require solutions for $P(t)$ that do not have singularities in the region within the range of the spherical polar coordinate θ (namely that it be nonsingular for the entire range $0 \le \theta \le \pi$, equivalent to $-1 \le t \le +1$). The solutions satisfying these conditions, called **associated Legendre functions**, are traditionally denoted P_l^m, with l a nonnegative integer. In Section 8.3 we discussed the Legendre equation as a 1-D eigenvalue problem, finding that the requirement of nonsingularity at $t = \pm 1$ is a sufficient boundary condition to make its solutions well defined. We found also that its eigenfunctions are the **Legendre polynomials** and that its eigenvalues

(λ in the present notation) have the values $l(l+1)$, where l is an integer. The generalization of these findings to the associated Legendre equation (that with nonzero m) shows that λ continues to be given as $l(l+1)$, but with the additional restriction that $l \geq |m|$. Details are deferred to Chapter 15.

Before continuing to the R equation, Eq. (9.78), let us observe that in deriving the Φ and Θ equations we have assumed that k^2 was a constant. However, if k^2 was not a constant, but an additive expression of the form

$$k^2 \quad \longrightarrow \quad f(r) + \frac{g(\theta)}{r^2} + \frac{h(\varphi)}{r^2 \sin^2 \theta},$$

we could still carry out the separation of variables, but the relatively familiar Φ and Θ equations we have identified will be changed in ways that make them different, and probably less tractable. However, if the departure of k^2 from a constant value is restricted to the form $k^2 = k^2(r)$, then the angular parts of the separation will remain as presented in Eqs. (9.74) and (9.79), and we only need to deal with increased generality in the R equation.

It is worth stressing that the great importance of this separation of variables in spherical polar coordinates stems from the fact that the case $k^2 = k^2(r)$ covers a tremendous amount of physics, such as a great deal of the theories of gravitation, electrostatics and atomic, nuclear, and particle physics. Problems with $k^2 = k^2(r)$ can be characterized as **central force problems**, and the use of spherical polar coordinates is natural in such problems. From both a practical and a theoretical point of view, it is a key observation that the angular dependence is isolated in Eqs. (9.74) and (9.77), or its equivalent, Eq. (9.79), that these equations are the same for all central force problems, and that **they can be solved exactly**. A detailed discussion of the angular properties of central force problems in quantum mechanics is deferred to Chapter 16.

Returning now to the remaining separated ODE, namely the R equation, we consider in some depth two special cases: (1) The case $k^2 = 0$, corresponding to the Laplace equation, and (2) k^2 a nonzero constant, corresponding to the Helmholtz equation. For both cases we assume that the Φ and Θ equations have been solved subject to the boundary conditions already discussed, so that the separation constant λ must have the value $l(l+1)$ for some nonnegative integer l. Continuing on the assumption that k^2 is a (possibly zero) constant, Eq. (9.79) expands into

$$r^2 R'' + 2r\, R' + \Big[k^2 r^2 - l(l+1)\Big] R = 0. \tag{9.80}$$

Taking first the case of the Laplace equation, for which $k^2 = 0$, Eq. (9.80) is easy to solve. Either by inspection or by attempting to carry out a series solution by the method of Frobenius, it is found that the initial term of the series, $a_0 r^s$, is by itself a complete solution to Eq. (9.80). In fact, substituting the assumed solution $R = r^s$ into Eq. (9.80), that equation reduces to

$$s(s-1)r^s + 2s\, r^s - l(l+1)r^s = 0,$$

showing that $s(s+1) = l(l+1)$, which has two solutions, $s = l$ (obviously), and $s = -l - 1$. In other words, given the value l from the choice of solution to the Θ equation, we find that the R equation (for the Laplace equation) has the two solutions r^l and r^{-l-1}, so its general solution takes the form

$$R(r) = A\,r^l + B\,r^{-l-1}. \tag{9.81}$$

Combining the solutions to the separated ODEs, and summing over all choices of the separation constants, we see that the most general solution of the Laplace equation that has a nonsingular angular dependence can be written

$$\psi(r,\theta,\varphi) = \sum_{l,m}(A_{lm}r^l + B_{lm}r^{-l-1})P_l^m(\cos\theta)(A'_{lm}\sin m\varphi + B'_{lm}\cos m\varphi). \tag{9.82}$$

If our problem now has Dirichlet or Neumann boundary conditions on a spherical surface (with the region under study either within or outside the sphere), we may be able (by methods more fully articulated in later chapters) to choose the coefficients in Eq. (9.82) so that the boundary conditions are satisfied. Note that if the region in which we are to solve the Laplace equation includes the origin, $r = 0$, then only the r^l term should be retained and we set B_{lm} to zero. If our region for the Laplace equation is, say, external to a sphere of some finite radius, then we must avoid the large-r divergence of r^l and set A_{lm} to zero, retaining only r^{-l-1}. More complicated cases, e.g., where we study the annular region between two concentric spheres, will require the retention of both A_{lm} and B_{lm} and will in general be somewhat more difficult.

We continue now to the case of nonzero but constant k^2. Equation (9.80) looks a lot like a Bessel equation, but differs therefrom by the coefficient "2" in the R' term and the factor k^2 that multiplies r^2 in the coefficient of R. Both these differences can be resolved by rewriting $R(r)$ as

$$R(r) = \frac{Z(kr)}{(kr)^{1/2}}, \tag{9.83}$$

which will then give us a differential equation for Z. Carrying out the differentiations to obtain R' and R'' in terms of Z, and changing the independent variable from r to $x = kr$, Eq. (9.83) becomes

$$x^2Z'' + xZ' + \left[x^2 - \left(l + \tfrac{1}{2}\right)^2\right]Z = 0, \tag{9.84}$$

showing that Z is a Bessel function, of order $l + \frac{1}{2}$. Returning to Eq. (9.83), we can now identify $R(r)$ in terms of quantities known as **spherical Bessel functions**, where $j_l(x)$, the spherical Bessel functions that are regular at $x = 0$, have definition

$$j_l(x) = \sqrt{\frac{\pi}{2x}}\,J_{l+1/2}(x).$$

Since the status of $R(r)$ as the solution to a homogeneous ODE is not affected by the scale factor in the definition of $j_l(x)$, we see that Eq. (9.83) is equivalent to the observation that Eq. (9.80) has a solution $j_l(kr)$. The spherical Bessel function that is the second solution of Eq. (9.80) is designated y_l, so that solution is $y_l(kr)$, and the general solution of Eq. (9.80) can be written

$$R(r) = Aj_l(kr) + By_l(kr). \tag{9.85}$$

We note here that the properties of spherical Bessel functions are discussed more fully in Chapter 14.

With the solutions to the radial ODE in hand, we can now write that the general solution to the Helmholtz equation in spherical polar coordinates takes the form

$$\psi(r,\theta,\varphi) = \sum_{l,m} \left[A_{lm}\, j_l(kr) + B_{lm}\, y_l(kr)\right] \times P_l^m(\cos\theta)(A'_{lm}\sin m\varphi + B'_{lm}\cos m\varphi). \tag{9.86}$$

The above discussion assumes that $k^2 > 0$; negative values of k^2 (and therefore imaginary values of k) simply correspond to our identifying an equation of the form $(\nabla^2 - k^2)\psi = 0$ as a somewhat peculiar case of $(\nabla^2 + k^2)\psi = 0$. For negative k^2, we can see we then get solutions that involve $j_l(kr)$ or $y_l(kr)$ with imaginary k. In order to avoid notations that unnecessarily involve imaginary quantities, it is usual to define a new set of functions $i_l(x)$ that are proportional to $j_l(ix)$, and are called **modified** spherical Bessel functions. The modified solutions parallel to $y_l(ix)$ are denoted $k_l(x)$. These functions are also discussed in Chapter 14.

The cases we have just surveyed do not, of course, cover all possibilities, and various other choices of $k^2(r)$ lead to problems that are of importance in physics. Without proceeding to a detailed analysis here, we cite a couple:

- Taking $k_2 = A/r + \lambda$ yields (with boundary condition that ψ vanish in the limit $r \to \infty$) the time-independent Schrödinger equation for the hydrogen atom; the R equation can then be identified as the associated Laguerre differential equation, discussed in Chapter 18.
- Taking $k^2 = Ar^2 + \lambda$ yields (with boundary condition at $r = \infty$) the equation for the 3-D quantum harmonic oscillator, for which the R equation can be reduced to the Hermite ODE, also discussed in Chapter 18.

Some other boundary-value problems lead to well-studied ODEs. However, sometimes the practicing physicist will encounter a radial equation that may have to be solved using the techniques presented in Chapter 7, or if all else fails, by numerical methods.

We close this subsection with an example that is a simple boundary-value problem in spherical coordinates.

Example 9.4.4 SPHERE WITH BOUNDARY CONDITION

In this example we solve the Laplace equation for the electrostatic potential $\psi(\mathbf{r})$ in a region interior to a sphere of radius a, using spherical polar coordinates (r,θ,φ) with origin at the center of the sphere. Our solution is to be subject to the Neumann boundary condition $d\psi/d\mathbf{n} = -V_0\cos\theta$ on the spherical surface. See Fig. 9.3.

To start, we note that totally arbitrary Neumann boundary conditions will not be consistent with our assumption of a charge-free sphere, as the integral of the normal derivative on the spherical surface gives, according to Gauss' law, a measure of the total charge within.

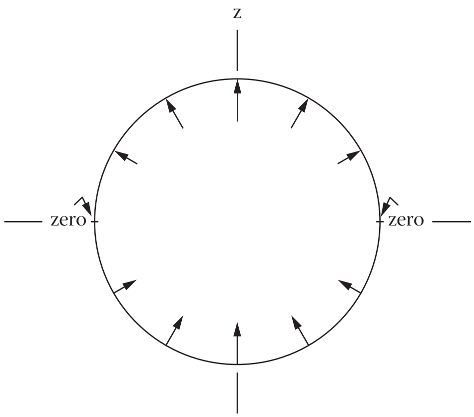

FIGURE 9.3 Arrows indicate sign and relative magnitude of the (inward) normal derivative of the electrostatic potential on a spherical surface (boundary condition for Example 9.4.4).

The present example is internally consistent, as

$$\int_S \cos\theta d\sigma = \int_0^\pi d\theta \int_0^{2\pi} d\varphi \cos\theta = 0.$$

Next, we need to take the general solution for the Laplace equation within a sphere, as given by Eq. (9.82), and calculate therefrom the inward normal derivative at $r = a$. Since the normal is in the $-r$ direction, we need only compute $-\partial\psi/\partial r$, evaluated at $r = a$. Noting that for the present problem $B_{lm} = 0$, our boundary condition becomes

$$-V\cos\theta = -\sum_{l,m} l\, A_{lm} a^{l-1} P_l^m(\cos\theta)(A'_{lm}\sin m\varphi + B'_{lm}\cos m\varphi).$$

Since the left-hand side of this equation is independent of φ, its right-hand side has nonzero coefficients only for $m = 0$, for which we only have the term originally containing B'_{l0}, because $\sin(0) = 0$. Thus, consolidating the constants, the boundary condition becomes the simpler form

$$-V\cos\theta = -\sum_l l\, A_l a^{l-1} P_l(\cos\theta), \tag{9.87}$$

Without having made a detailed study of the properties of Legendre functions, the solution of an equation of this type might need to be deferred to Chapter 15, but this one is easy to solve because $P_1(\cos\theta) = \cos\theta$ (see Legendre polynomials in Table 15.1) Thus, from Eq. (9.87),

$$l A_l a^{l-1} = V\delta_{l1},$$

so $A_1 = V$ and all the other coefficients except A_0 vanish. The coefficient A_0 is not determined by the boundary conditions and represents an arbitrary constant that may be added to the potential. Thus, the potential within the sphere has the form

$$\psi = V r P_1(cos\theta) + A_0 = V r \cos\theta + A_0 = V z + A_0,$$

corresponding to a uniform electric field within the sphere, in the $-z$ direction and of magnitude V. The electric field is, of course, unaffected by the arbitrary value of the constant A_0. ■

Summary: Separated-Variable Solutions

For convenient reference, the forms of the solutions of Laplace's and Helmholtz's equations for spherical polar coordinates are collected in Table 9.2. Although the ODEs obtained from the separation of variables are the same irrespective of the boundary conditions, the ODE solutions to be used, and the constants of separation, do depend on the boundaries. Boundaries with less than spherical symmetry may lead to values of m and l that are not integral, and may also require use of the second solution of the Legendre equation (quantities normally denoted Q_l^m). Engineering applications frequently require solutions to PDEs for regions of low symmetry, but such problems are nowadays almost universally approached using numerical, rather than analytical methods. Consequently, Table 9.2 only contains data that are relevant for problems inside or outside a spherical boundary, or between two concentric spherical boundaries. This restriction to spherical symmetry causes the angular portion of the solutions to be uniquely of the form we have already identified.

In contrast to the unique angular solution, both linearly independent solutions to the radial ODE are relevant, with the choice of solution dependent on the geometry. Solutions within a sphere must employ only the radial functions that are regular at the origin, i.e., r^l, j_l, or i_l. Solutions external to a sphere may employ r^{-l-1}, k_l (defined so that it will decay exponentially to zero at large r), or a linear combination of j_l and y_l (both of which are oscillatory and decay as $r^{-1/2}$). Solutions between concentric spheres can use both the radial functions appropriate to the PDE.

It is also possible to summarize the forms of solution to the Laplace and Helmholtz equations in circular cylindrical coordinates, if we restrict attention to problems that have circular symmetry about the axial direction of the coordinate system. However, the situation is considerably more complicated than for spherical coordinates, as we now have two

Table 9.2 Solutions of PDEs in Spherical Polar Coordinates[a]

$$\psi = \sum_{l,m} f_l(r) P_l^m(\cos\theta) \begin{Bmatrix} a_{lm}\cos m\varphi + b_{lm}\sin m\varphi) \\ \text{or} \\ c_{lm} e^{im\varphi} \end{Bmatrix}$$

$\nabla^2\psi = 0$	$f_l(r) = r^l,\ r^{-l-1}$
$\nabla^2\psi + k^2\psi = 0$	$f_l(r) = j_l(kr),\ y_l(kr)$
$\nabla^2\psi - k^2\psi = 0$	$f_l(r) = i_l(kr),\ k_l(kr)$

[a] For i_l, j_l, k_l, y_l, see Chapter 14; for P_l^m, see Chapters 15 and 16.

Table 9.3 Solutions of PDEs in Circular Cylindrical Coordinates[a]

$$\psi = \sum_{m,\alpha} f_{m\alpha}(\rho) g_\alpha(z) \begin{Bmatrix} a_{m\alpha}\cos m\varphi + b_{m\alpha}\sin m\varphi) \\ \text{or} \\ c_{m\alpha}e^{im\varphi} \end{Bmatrix}$$

$\nabla^2\psi = 0$		$f_{m\alpha}(\rho) = J_m(\alpha\rho), Y_m(\alpha\rho)$	$g_\alpha(z) = e^{\alpha z},\ e^{-\alpha z}$
	or	$f_{m\alpha}(\rho) = I_m(\alpha\rho), K_m(\alpha\rho)$	$g_\alpha(z) = \sin(\alpha z), \cos(\alpha z)$ or $e^{i\alpha z}$
	or	$f_{m\alpha}(\rho) = \rho^m,\ \rho^{-m}$	$g_\alpha(z) = 1$
$\nabla^2\psi + \lambda\psi = 0$		$f_{m\alpha}(\rho) = J_m(\alpha\rho), Y_m(\alpha\rho)$	
		if $\beta^2 = \alpha^2 - \lambda > 0$,	$g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$
		if $\beta^2 = \lambda - \alpha^2 > 0$,	$g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$
		if $\lambda = \alpha^2$,	$g_\alpha(z) = 1$
	or	$f_{m\alpha}(\rho) = I_m(\alpha\rho), K_m(\alpha\rho)$	
		if $\beta^2 = -\lambda - \alpha^2 > 0$,	$g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$
		if $\beta^2 = \lambda + \alpha^2 > 0$,	$g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$
		if $\lambda = -\alpha^2$,	$g_\alpha(z) = 1$
	or	$f_{m\alpha}(\rho) = \rho^m,\ \rho^{-m}$	
		if $\beta^2 = -\lambda > 0$,	$g_\alpha(z) = e^{\beta z},\ e^{-\beta z}$
		if $\beta^2 = \lambda > 0$,	$g_\alpha(z) = \sin(\beta z), \cos(\beta z)$ or $e^{i\beta z}$

[a] The parameter α can have any real values consistent with the boundary conditions. For I_m, J_m, K_m, Y_m, see Chapter 14.

coordinates (ρ and z) that can have a variety of boundary conditions, in contrast to the single such coordinate (r) in the spherical system. In spherical coordinates the form of the radial function is completely determined by the PDE, and specific problems differ only in the choice (or relative weight) of the two linearly independent radial solutions. But in cylindrical coordinates the forms of the ρ and z solutions, as well as their coefficients, are determined by the boundary conditions, and not entirely by the value of the constant in the Helmholtz equation. Choices of the ρ and z solutions, though coupled, can vary widely. For details, the reader is referred to Table 9.3.

Our final observations of this section deal with the functions we encountered in the course of the separations in cylindrical and spherical coordinates. For the purpose of this discussion, it is useful to think of our PDE as an operator equation subject to boundary conditions. If, in cylindrical coordinates, we restrict attention to PDEs in which the parameter k^2 is independent of φ (and with boundary conditions that do not depend upon φ), we have chosen our operator equation as one that has circular symmetry. Moreover, we will then always get the same Φ equation, with (of course) the same solutions. In these circumstances, the solutions will have symmetry properties derived from those of our overall boundary-value problem.[6] The Φ equation can also be thought of as an operator equation, and we can go further and identify the operator as $L_z^2 = -\partial^2/\partial\varphi^2$, where L_z is the z component of the angular momentum. The solutions of the Φ equation are eigenfunctions of this operator; the reason they can occur as part of the PDE solution is because

[6]Note that the solutions to a boundary-value problem need not have the full problem symmetry (a point that will be elaborated in great detail when we develop group-theoretical methods). An obvious example is that the Sun-Earth gravitational potential is spherically symmetric, while the most familiar solution (the Earth's orbit) is planar. The dilemma is resolved by noting that the spherical symmetry manifests itself in the possible existence of Earth orbits at all angular orientations.

L_z^2 commutes with the operator defining the PDE (clearly so, because the PDE operator does not contain φ). In other words, because L_z^2 and the PDE operator commute, they will have simultaneous eigenfunctions, and the overall solutions of the PDE can be labeled to identify the L_z^2 eigenfunction that was chosen.

Looking now at the situation in spherical polar coordinates, we note that if k^2 is independent of the angles, i.e., $k^2 = k^2(r)$, then our PDE always has the same angular solutions $\Theta_{lm}(\theta)\Phi_m(\varphi)$. Looking further at the angular terms of our PDE, we can identify them as the operator L^2, and we see that the angular solutions we have found are eigenfunctions of this operator. When the PDE operator is independent of the angles, it will commute with L^2 and the solutions to the PDE can be labeled accordingly. These symmetry features are very important and are discussed in great detail in Chapter 16.

Exercises

9.4.1 By letting the operator $\nabla^2 + k^2$ act on the general form $a_1\psi_1(x, y, z) + a_2\psi_2(x, y, z)$, show that it is linear, i.e., that $(\nabla^2 + k^2)(a_1\psi_1 + a_2\psi_2) = a_1(\nabla^2 + k^2)\psi_1 + a_2(\nabla^2 + k^2)\psi_2$.

9.4.2 Show that the Helmholtz equation,

$$\nabla^2\psi + k^2\psi = 0,$$

is still separable in circular cylindrical coordinates if k^2 is generalized to $k^2 + f(\rho) + (1/\rho^2)g(\varphi) + h(z)$.

9.4.3 Separate variables in the Helmholtz equation in spherical polar coordinates, splitting off the radial dependence **first**. Show that your separated equations have the same form as Eqs. (9.74), (9.77), and (9.78).

9.4.4 Verify that

$$\nabla^2\psi(r, \theta, \varphi) + \left[k^2 + f(r) + \frac{1}{r^2}g(\theta) + \frac{1}{r^2\sin^2\theta}h(\varphi)\right]\psi(r, \theta, \varphi) = 0$$

is separable (in spherical polar coordinates). The functions f, g, and h are functions only of the variables indicated; k^2 is a constant.

9.4.5 An atomic (quantum mechanical) particle is confined inside a rectangular box of sides a, b, and c. The particle is described by a wave function ψ that satisfies the Schrödinger wave equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi.$$

The wave function is required to vanish at each surface of the box (but not to be identically zero). This condition imposes constraints on the separation constants and therefore on the energy E. What is the smallest value of E for which such a solution can be obtained?

$$ANS.\quad E = \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right).$$

9.4.6 The quantum mechanical angular momentum operator is given by $\mathbf{L} = -i(\mathbf{r} \times \boldsymbol{\nabla})$. Show that

$$\mathbf{L} \cdot \mathbf{L}\psi = l(l+1)\psi$$

leads to the associated Legendre equation.
Hint. Section 8.3 and Exercise 8.3.1 may be helpful.

9.4.7 The 1-D Schrödinger wave equation for a particle in a potential field $V = \frac{1}{2}kx^2$ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi(x).$$

(a) Defining

$$a = \left(\frac{mk}{\hbar^2}\right)^{1/4}, \quad \lambda = \frac{2E}{\hbar}\left(\frac{m}{k}\right)^{1/2},$$

and setting $\xi = ax$, show that

$$\frac{d^2\psi(\xi)}{d\xi^2} + (\lambda - \xi^2)\psi(\xi) = 0.$$

(b) Substituting

$$\psi(\xi) = y(\xi)e^{-\xi^2/2},$$

show that $y(\xi)$ satisfies the Hermite differential equation.

9.5 Laplace and Poisson Equations

The Laplace equation can be considered the prototypical elliptic PDE. At this point we supplement the discussion motivated by the method of separation of variables with some additional observations. The importance of Laplace's equation for electrostatics has stimulated the development of a great variety of methods for its solution in the presence of boundary conditions ranging from simple and symmetrical to complicated and convoluted. Techniques for present-day engineering problems tend to rely heavily on computational methods. The thrust of this section, however, will be on general properties of the Laplace equation and its solutions.

The basic properties of the Laplace equation are independent of the coordinate system in which it is expressed; we assume for the moment that we will use Cartesian coordinates. Then, because the PDE sets the sum of the second derivatives, $\partial^2\psi/\partial x_i^2$, to zero, it is obvious that if any of the second derivatives has a positive sign, at least one of the others must be negative. This point is illustrated in Example 9.4.1, where the x and y dependence of a solution to the Laplace equation was sinusoidal, and as a result, the z dependence was exponential (corresponding to different signs for the second derivative). Since the second derivative is a measure of curvature, we conclude that if ψ has positive curvature in any coordinate direction, it must have negative curvature in some other coordinate direction. That observation, in turn, means that all the **stationary points** of ψ (points where its first derivatives in all directions vanish) must be **saddle points**, not maxima or minima.

Since the Laplace equation describes the static electric potential in charge-free regions, we conclude that the potential cannot have an extremum at a point where there is no charge. A corollary to this observation is that the extrema of the electrostatic potential in a charge-free region must be on the boundary of the region.

A related property of the Laplace equation is that its solution, subject to Dirichlet boundary conditions for the entire closed boundary of its region, is unique. This property applies also to its inhomogeneous generalization, the Poisson equation. The proof is simple: Suppose there are two distinct solutions ψ_1 and ψ_2 for the same boundary conditions. Then, their difference $\psi = \psi_1 - \psi_2$ (for either the Laplace or Poisson equation) will be a solution to the Laplace equation with $\psi = 0$ on the boundary. Since ψ cannot have extrema within the bounded region, it must be zero everywhere, meaning that $\psi_1 = \psi_2$.

If we have a Laplace or Poisson equation subject to Neumann boundary conditions on the entire closed boundary of its region, then the difference $\psi = \psi_1 - \psi_2$ of two solutions will also be a solution to the Laplace equation with a zero Neumann boundary condition. To analyze this situation, we invoke Green's Theorem, in the form provided by Eq. (3.86), taking both u and v of that equation to be ψ. Equation (3.86) then becomes

$$\int_S \psi \frac{\partial \psi}{\partial \mathbf{n}} dS = \int_V \psi \nabla^2 \psi \, d\tau + \int_V \nabla \psi \cdot \nabla \psi \, d\tau. \tag{9.88}$$

The boundary condition causes the left-hand side of Eq. (9.88) to vanish, the first integral on the right-hand side vanishes because ψ is a solution of the Laplace equation, and the remaining integral on the right-hand side must therefore also vanish. But that integral can only vanish if $\nabla\psi$ is zero everywhere, which can only be true if ψ is constant. Thus, solutions to the Laplace equation with Neumann boundary conditions are also unique, except for an additive constant to the potential.

An oft-cited application of this uniqueness theorem is the solution of electrostatics problems by the method of images, which replaces a problem containing boundaries by one without a boundary but with additional charge added in such a way that the potential at the boundary location has the desired value. For example, a positive charge in front of a grounded boundary (one with $\psi = 0$) can be augmented by a negative charge at the mirror-image position behind the boundary. Then the two-charge system (ignoring the boundary) will yield the desired zero potential at the boundary location, and the uniqueness theorem tells us that the potential calculated for the two-charge system must be the same (within the original region) as that for the original system.

Exercises

9.5.1 Verify that the following are solutions of Laplace's equation:

$$\text{(a)} \quad \psi_1 = 1/r, \; r \neq 0, \quad \text{(b)} \quad \psi_2 = \frac{1}{2r} \ln \frac{r+z}{r-z}.$$

9.5.2 If Ψ is a solution of Laplace's equation, $\nabla^2 \Psi = 0$, show that $\partial\Psi/\partial z$ is also a solution.

9.5.3 Show that an argument based on Eq. (9.88) can be used to prove that the Laplace and Poisson equations with Dirichlet boundary conditions have unique solutions.

9.6 WAVE EQUATION

The wave equation is the prototype hyperbolic PDE. As we have seen earlier in this chapter, hyperbolic PDEs have two *characteristics*, and for the equation

$$\frac{1}{c^2}\frac{\partial^2\psi}{\partial t^2} = \frac{\partial^2\psi}{\partial x^2}, \tag{9.89}$$

the characteristics are lines of constant $x - ct$ and those of constant $x + ct$. This means that the general solution to Eq. (9.89) takes the form

$$\psi(x, t) = f(x - ct) + g(x + ct), \tag{9.90}$$

with f and g completely arbitrary.

Viewing x as a position variable and t as the time, we can interpret $f(x - ct)$ as a wave, moving with velocity c, in the $+x$ direction. By this we mean that the entire profile of f, as a function of x at $t = 0$, will be shifted uniformly toward positive x by an amount c when $t = 1$. See Fig. 9.4. Similarly, $g(x + ct)$ describes a wave moving at velocity c in the $-x$ direction. Because f and g are arbitrary, the **traveling waves** they describe need not be sinusoidal or periodic, but may be entirely irregular; moreover, there is no requirement that f and g have any particular relationship to each other.

An obvious special case of the general situation described above is that when $f(x - ct)$ is chosen to be sinusoidal, $f = \sin(x - ct)$. For simplicity we have taken f to have unit amplitude and wavelength 2π. We also take $g(x + ct)$ to be $g = \sin(x + ct)$, a sinusoidal wave of the same wavelength and amplitude traveling in the direction opposite to f. At a point x and time t, these two waves add to produce a resultant

$$\psi(x, t) = \sin(x - ct) + \sin(x + ct),$$

which, using trigonometric identities, can be rearranged to

$$\psi(x, t) = (\sin x \cos ct - \cos x \sin ct) + (\sin x \cos ct + \cos x \sin ct) = 2 \sin x \cos ct.$$

This form for ψ can be identified as a **standing wave** distribution, meaning that the time evolution of the wave's profile in x is an oscillation in amplitude, with the wave pattern not moving in either direction. An obvious point of difference from a traveling wave is that for a standing wave, the **nodes** (points where $\psi = 0$) are stationary in time, while in a traveling wave they are moving in time at velocity $\pm c$.

Our current interest in traveling vs. standing waves is their relation to solutions to the wave equation that we might find using the method of separation of variables. That method would obviously lead us to standing-wave solutions. However, it is useful to note that the totality of the solution set from the separated variables has the same content as

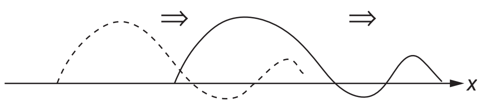

FIGURE 9.4 Traveling wave $f(x - ct)$. Dashed line is profile at $t = 0$; full line is profile at a time $t > 0$.

the traveling-wave solutions. For example, the products $\sin x \cos ct$ and $\cos x \sin ct$ are solutions we would get by separating the variables, and linear combinations of these yield $\sin(x \pm ct)$.

d'Alembert's Solution

While all ways of writing the general solution to the wave equation are mathematically equivalent, diverse forms differ in their convenience of use for various purposes. To illustrate this, we consider how we might construct a solution to the wave equation, given, as an initial condition, (1) the entire spatial distribution of the wave amplitude at $t = 0$ and (2) the time derivative of the wave amplitude at $t = 0$ for the entire spatial distribution. The solution to this problem is generally referred to as **d'Alembert's solution** of the wave equation; it was also (and slightly earlier) found by Euler.

We start by using Eq. (9.90) to write our initial conditions in terms of the presently unknown functions f and g:

$$\psi(x, 0) = f(x) + g(x), \tag{9.91}$$

$$\left.\frac{\partial \psi(x,t)}{\partial t}\right|_{t=0} = -cf'(x) + cg'(x). \tag{9.92}$$

We now integrate Eq. (9.92) between the limits $x - ct$ and $x + ct$ (and divide the result by $2c$), obtaining

$$\frac{1}{2c}\int_{x-ct}^{x+ct} \frac{\partial \psi(x,0)}{\partial t} dx = \frac{1}{2}\big[-f(x+ct) + f(x-ct) + g(x+ct) - g(x-ct)\big]. \tag{9.93}$$

From Eq. (9.91), we also have

$$\frac{1}{2}\big[\psi(x+ct,0) + \psi(x-ct,0)\big] = \frac{1}{2}\big[f(x+ct) + g(x+ct) + f(x-ct) + g(x-ct)\big]. \tag{9.94}$$

Adding together the right-hand sides of Eqs. (9.93) and (9.94), half the terms cancel, and those that survive combine to give the result

$$f(x-ct) + g(x+ct), \quad \text{which is } \psi(x,t).$$

Therefore, from the left-hand sides of Eqs. (9.93) and (9.94), we obtain the final result

$$\psi(x,t) = \frac{1}{2}\big[\psi(x+ct,0) + \psi(x-ct,0)\big] + \frac{1}{2c}\int_{x-ct}^{x+ct} \frac{\partial \psi(x,0)}{\partial t} dx. \tag{9.95}$$

This equation gives $\psi(x,t)$ in terms of data at $t = 0$ that are within the distance ct of the point x. This is a reasonable result, since ct is the distance that waves in this problem can move between times $t = 0$ and $t = t$. More specifically, Eq. (9.95) contains terms that represent half the $t = 0$ amplitude at distances $\pm ct$ from x (half, because a disturbance that starts at these points is split between propagation in both directions), plus an additional integral that accumulates the effect of the initial amplitude derivative over the region of influence.

Exercises

Solve the wave equation, Eq. (9.89), subject to the indicated conditions.

9.6.1 Determine $\psi(x,t)$ given that at $t=0$ $\psi_0(x)=\sin x$ and $\partial\psi(x)/\partial t=\cos x$.

9.6.2 Determine $\psi(x,t)$ given that at $t=0$ $\psi_0(x)=\delta(x)$ (Dirac delta function) and the initial time derivative of ψ is zero.

9.6.3 Determine $\psi(x,t)$ given that at $t=0$ $\psi_0(x)$ is a single square-wave pulse as defined below, and the initial time derivative of ψ is zero.

$$\psi_0(x)=0,\quad |x|>a/2,\qquad \psi_0(x)=1/a,\quad |x|<a/2.$$

9.6.4 Determine $\psi(x,t)$ given that at $t=0$ $\psi_0=0$ for all x, but $\partial\psi/\partial t=\sin(x)$.

9.7 Heat-Flow, or Diffusion PDE

Here we return to a parabolic PDE to develop methods that adapt a special solution of a PDE to boundary conditions by introducing parameters. The methods are fairly general and apply to other second-order PDEs with constant coefficients as well. To some extent, they are complementary to the earlier basic separation method for finding solutions in a systematic way.

We consider the 3-D time-dependent diffusion PDE for an isotropic medium, using it to describe heat flow subject to given boundary conditions. Assuming isotropy actually is not much of a restriction because, in case we have different (constant) rates of diffusion in different directions, for example, in wood, our heat-flow PDE takes the form

$$\frac{\partial\psi}{\partial t}=a^2\frac{\partial^2\psi}{\partial x^2}+b^2\frac{\partial^2\psi}{\partial y^2}+c^2\frac{\partial^2\psi}{\partial z^2}, \tag{9.96}$$

if we put the coordinate axes along the principal directions of anisotropy. Now we simply rescale the coordinates using the substitutions $x=a\xi$, $y=b\eta$, $z=c\zeta$ to get back the original isotropic form of Eq. (9.96),

$$\frac{\partial\Phi}{\partial t}=\frac{\partial^2\Phi}{\partial\xi^2}+\frac{\partial^2\Phi}{\partial\eta^2}+\frac{\partial^2\Phi}{\partial\zeta^2}, \tag{9.97}$$

for the temperature distribution function $\Phi(\xi,\eta,\zeta,t)=\psi(x,y,z,t)$.

For simplicity, we first solve the time-dependent PDE for a homogeneous one-dimensional medium, a long metal rod in the x-direction, for which the PDE is

$$\frac{\partial\psi}{\partial t}=a^2\frac{\partial^2\psi}{\partial x^2}, \tag{9.98}$$

where the constant a measures the diffusivity, or heat conductivity, of the medium. We obtain solutions to this linear PDE with constant coefficients by the method of separation of variables, for which we set $\psi(x,t)=X(x)T(t)$, leading to the separate equations

$$\frac{1}{T}\frac{dT}{dt}=\beta,\qquad \frac{1}{X}\frac{d^2X}{dx^2}=\frac{\beta}{a^2}.$$

These equations have, for any nonzero value of β, solutions $T=e^{\beta t}$ and $X=e^{\pm\alpha x}$, with $\alpha^2=\beta/a^2$. We seek solutions whose time dependence decays exponentially at large t, that is, solutions with negative values of β, and therefore set $\alpha = i\omega$, $a^2 = -\omega^2$ for real ω, and have

$$\psi(x,t) = e^{i\omega x} e^{-\omega^2 a^2 t} = (\cos\omega x \pm i \sin\omega x) e^{-\omega^2 a^2 t}. \tag{9.99}$$

Note that $\beta = 0$, for which

$$\psi(x,t) = C_0' x + C_0, \tag{9.100}$$

is also included in the solution set for the PDE. If we use this solution for a rod of infinite length, we must set $C_0' = 0$ to avoid a nonphysical divergence; in any case, the value of C_0 is then the constant value that the temperature approaches at long times.

Forming real linear combinations of $\sin\omega x$ and $\cos\omega x$ with arbitrary coefficients, and keeping the $\beta = 0$ solution, we obtain from Eq. (9.99) for any choice of A, B, ω, C_0', and C_0, a solution

$$\psi(x,t) = (A\cos\omega x + B\sin\omega x) e^{-\omega^2 a^2 t} + C_0' x + C_0. \tag{9.101}$$

Solutions for different values of these parameters can now be combined as needed to form an overall solution consistent with the required boundary conditions.

If the rod we are studying is finite in length, it may be that the boundary conditions can be satisfied if we restrict ω to discrete nonzero values that are multiples of a basic value ω_0. For a rod of infinite length, it may be better to let ω assume a continuous range of values, so that $\psi(x,t)$ will have the general form

$$\psi(x,t) = \int [A(\omega)\cos\omega x + B(\omega)\sin\omega x] e^{-a^2\omega^2 t} d\omega + C_0. \tag{9.102}$$

We call specific attention to the fact that

- Forming linear combinations of solutions by summation or integration over parameters is a powerful and standard method for generalizing specific PDE solutions in order to adapt them to boundary conditions.

Example 9.7.1 A Specific Boundary Condition

Let us solve a 1-D case explicitly, where the temperature at time $t = 0$ is $\psi_0(x) = 1 =$ constant in the interval between $x = +1$ and $x = -1$ and zero for $x > 1$ and $x < -1$. At the ends, $x = \pm 1$, the temperature is always held at zero. Note that this problem, including its initial conditions, has even parity, $\psi_0(x) = \psi_0(-x)$, so $\psi(x,t)$ must also be even.

We choose the spatial solutions of Eq. (9.98) to be of the form given in Eq. (9.101), but restricted to $C_0' = C_0 = 0$ (since the $t \to \infty$ limit of $\psi(x,t)$ is zero for the entire range $-1 \le x \le 1$), and to $\cos(l\pi x/2)$ for odd integer l, because these functions are the even-parity members of an orthonormal basis for the interval $-1 \le x \le 1$ that satisfy the boundary condition $\psi = 0$ at $x = \pm 1$. Then, at $t = 0$ our solution takes the form

$$\psi(x,0) = \sum_{l=1}^{\infty} a_l \cos\frac{\pi l x}{2}, \quad -1 < x < 1,$$

and we need to choose the coefficients a_l so that $\psi(x,0) = 1$.

Using the orthonormality, we compute

$$a_l = \int_{-1}^{1} 1 \cdot \cos\frac{\pi l x}{2} = \frac{2}{l\pi}\sin\frac{\pi l x}{2}\Big|_{x=-1}^{1}$$

$$= \frac{4}{\pi l}\sin\frac{l\pi}{2} = \frac{4(-1)^m}{(2m+1)\pi}, \qquad l = 2m+1.$$

Including its time dependence, the full solution is given by the series

$$\psi(x,t) = \frac{4}{\pi}\sum_{m=0}^{\infty}\frac{(-1)^m}{2m+1}\cos\left[(2m+1)\frac{\pi x}{2}\right]e^{-t((2m+1)\pi a/2)^2}, \tag{9.103}$$

which converges absolutely for $t > 0$ but only conditionally at $t = 0$, as a result of the discontinuity at $x = \pm 1$. ■

We are now ready to consider the diffusion equation in three dimensions. We start by assuming a solution of the form $\psi = f(x, y, z)T(t)$, and separate the spatial from the time dependence. As in the 1-D case, $T(t)$ will have exponentials as solutions, and we can choose the solution that decays exponentially at large t. Assigning the separation constant the value $-k^2$, so that the time dependence is $\exp(-k^2 t)$, the separated equation in the spatial coordinates takes the form

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} + k^2 f = 0, \tag{9.104}$$

which we recognize as the **Helmholtz** equation. Assuming that we can solve this equation for various values of k^2 by further separations of variables or by other means, we can form whatever sum or integral of individual solutions that may be needed to satisfy the boundary conditions.

Alternate Solutions

In an alternative approach to the heat flow equation, we now return to the one-dimensional PDE, Eq. (9.98), seeking solutions of a new functional form $\psi(x,t) = u(x/\sqrt{t})$, which is suggested by dimensional considerations and experimental data. Substituting $u(\xi)$, $\xi = x/\sqrt{t}$, into Eq. (9.98) using

$$\frac{\partial \psi}{\partial x} = \frac{u'}{\sqrt{t}}, \qquad \frac{\partial^2 \psi}{\partial x^2} = \frac{u''}{t}, \qquad \frac{\partial \psi}{\partial t} = -\frac{x}{2\sqrt{t^3}}u' \tag{9.105}$$

with the notation $u'(\xi) \equiv du/d\xi$, the PDE is reduced to the ODE

$$2a^2 u''(\xi) + \xi u'(\xi) = 0. \tag{9.106}$$

Writing this ODE as

$$\frac{u''}{u'} = -\frac{\xi}{2a^2},$$

we can integrate it once to get $\ln u' = -\xi^2/4a^2 + \ln C_1$, where C_1 is an integration constant. Exponentiating and integrating again we find the general solution

$$u(\xi) = C_1 \int_0^{\xi} e^{-\xi^2/4a^2} d\xi + C_2, \tag{9.107}$$

which contains two integration constants C_i. We initialize this solution at time $t = 0$ to temperature $+1$ for $x > 0$ and -1 for $x < 0$, corresponding to $u(\infty) = +1$ and $u(-\infty) = -1$. Noting that

$$\int_0^{\infty} e^{-\xi^2/4a^2} d\xi = a\sqrt{\pi},$$

a case of the integral evaluated in Eq. (1.148), we obtain

$$u(\infty) = a\sqrt{\pi}C_1 + C_2 = 1, \quad u(-\infty) = -a\sqrt{\pi}C_1 + C_2 = -1,$$

which fixes the constants $C_1 = 1/a\sqrt{\pi}$, $C_2 = 0$. We therefore have the specific solution

$$\psi = \frac{1}{a\sqrt{\pi}} \int_0^{x/\sqrt{t}} e^{-\xi^2/4a^2} d\xi = \frac{2}{\sqrt{\pi}} \int_0^{x/2a\sqrt{t}} e^{-v^2} dv = \operatorname{erf}\left(\frac{x}{2a\sqrt{t}}\right), \tag{9.108}$$

where **erf** is the standard name for Gauss' error function (one of the special functions listed in Table 1.2). We need to generalize this specific solution to adapt it to boundary conditions.

To this end we now generate **new solutions of the PDE with constant coefficients by differentiating the special solution** given in Eq. (9.108). In other words, if $\psi(x,t)$ solves the PDE in Eq. (9.98), so do $\partial\psi/\partial t$ and $\partial\psi/\partial x$, because these derivatives and the differentiations of the PDE commute; that is, the order in which they are carried out does not matter. Note carefully that this method no longer works if any coefficient of the PDE depends on t or x explicitly. However, PDEs with constant coefficients dominate in physics. Examples are Newton's equations of motion in classical mechanics, the wave equations of electrodynamics, and Poisson's and Laplace's equations in electrostatics and gravity. Even Einstein's nonlinear field equations of general relativity take on this special form in local geodesic coordinates.

Therefore, by differentiating Eq. (9.108) with respect to x, we find the simpler, more basic solution,

$$\psi_1(x,t) = \frac{1}{a\sqrt{t\pi}} e^{-x^2/4a^2 t}, \tag{9.109}$$

and, repeating the process, another basic solution

$$\psi_2(x,t) = \frac{x}{2a^3\sqrt{t^3\pi}} e^{-x^2/4a^2 t}. \tag{9.110}$$

Again, these solutions have to be generalized to adapt them to boundary conditions. And there is yet another method of generating new solutions of a PDE with constant coefficients: We can **translate** a given solution, for example, $\psi_1(x,t) \to \psi_1(x-\alpha,t)$, and then

integrate over the translation parameter α. Therefore,

$$\psi(x,t) = \frac{1}{2a\sqrt{t\pi}} \int_{-\infty}^{\infty} C(\alpha) e^{-(x-\alpha)^2/4a^2t} d\alpha \tag{9.111}$$

is again a solution, which we rewrite using the substitution

$$\xi = \frac{x-\alpha}{2a\sqrt{t}}, \quad \alpha = x - 2a\xi\sqrt{t}, \quad d\alpha = -2a\sqrt{t}d\xi. \tag{9.112}$$

These substitutions lead to

$$\psi(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} C(x - 2a\xi\sqrt{t}) e^{-\xi^2} d\xi, \tag{9.113}$$

a solution of our PDE. Equation (9.113) is in a form permitting us to understand the significance of the weight function $C(x)$ from the translation method. If we set $t = 0$ in that equation, the function C in the integrand then becomes independent of ξ, and the integral can then be recognized as

$$\int_{-\infty}^{\infty} e^{-\xi^2} d\xi = \sqrt{\pi},$$

a well-known result equivalent to Eq. (1.148). Equation (9.113) then becomes the simpler form

$$\psi(x,0) = C(x), \quad \text{or} \quad C(x) = \psi_0(x),$$

where ψ_0 is the initial spatial distribution of ψ. Using this notation, we can write the solution to our PDE as

$$\psi(x,t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \psi_0(x - 2a\xi\sqrt{t}) e^{-\xi^2} d\xi, \tag{9.114}$$

a form that explicitly displays the role of the boundary (initial) condition. From Eq. (9.114) we see that the initial temperature distribution, $\psi_0(x)$, spreads out over time and is damped by the Gaussian weight function.

Example 9.7.2 Special Boundary Condition Again

We consider now a problem similar to Example 9.7.1, but instead of keeping $\psi = 0$ at all times at $x = \pm 1$, we regard the system as infinite in length, with $\psi_0 = 0$ everywhere except for $|x| < 1$, where $\psi_0 = 1$. This change makes Eq. (9.114) usable, because our PDE now applies over the range $(-\infty, \infty)$, and heat will flow (and temporarily increase the temperature) at and beyond $|x| = 1$.

The range of $\psi_0(x)$ corresponds to a range of ξ with endpoints found from $x - 2a\xi\sqrt{t} = \pm 1$, so our solution becomes

$$\psi(x,t) = \frac{1}{\sqrt{\pi}} \int_{(x-1)/2a\sqrt{t}}^{(x+1)/2a\sqrt{t}} e^{-\xi^2} d\xi.$$

In terms of the error function, we can also write this solution as

$$\psi(x,t) = \frac{1}{2}\left[\operatorname{erf}\left(\frac{x+1}{2a\sqrt{t}}\right) - \operatorname{erf}\left(\frac{x-1}{2a\sqrt{t}}\right)\right]. \tag{9.115}$$

Equation (9.115) applies for all x, including $|x| > 1$. ■

Next we consider the problem of heat flow for an extended **spherically symmetric** medium centered at the origin, suggesting that we should use polar coordinates r, θ, φ. We expect a solution of the form $u(r,t)$. Using Eq. (3.158) for the Laplacian, we find the PDE

$$\frac{\partial u}{\partial t} = a^2\left(\frac{\partial^2 u}{\partial r^2} + \frac{2}{r}\frac{\partial u}{\partial r}\right), \tag{9.116}$$

which we transform to the 1-D heat-flow PDE by the substitution

$$u = \frac{v(r,t)}{r}, \quad \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial r} - \frac{v}{r^2}, \quad \frac{\partial u}{\partial t} = \frac{1}{r}\frac{\partial v}{\partial t},$$
$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{r}\frac{\partial^2 v}{\partial r^2} - \frac{2}{r^2}\frac{\partial v}{\partial r} + \frac{2v}{r^3}. \tag{9.117}$$

This yields the PDE

$$\frac{\partial v}{\partial t} = a^2\frac{\partial^2 v}{\partial r^2}. \tag{9.118}$$

Example 9.7.3 SPHERICALLY SYMMETRIC HEAT FLOW

Let us apply the 1-D heat-flow PDE to a spherically symmetric heat flow under fairly common boundary conditions, where x is replaced by the radial variable. Initially we have zero temperature everywhere. Then, at time $t = 0$, a finite amount of heat energy Q is released at the origin, spreading evenly in all directions. What is the resulting spatial and temporal temperature distribution?

Inspecting our special solution in Eq. (9.110) we see that, for $t \to 0$, the temperature

$$\frac{v(r,t)}{r} = \frac{C}{\sqrt{t^3}}e^{-r^2/4a^2t} \tag{9.119}$$

goes to zero for all $r \neq 0$, so zero initial temperature is guaranteed. As $t \to \infty$, the temperature $v/r \to 0$ for all r including the origin, which is implicit in our boundary conditions. The constant C can be determined from energy conservation, which gives (for arbitrary t) the constraint

$$Q = \sigma\rho\int\frac{v}{r}d\tau = \frac{4\pi\sigma\rho C}{\sqrt{t^3}}\int_0^\infty r^2 e^{-r^2/4a^2t}dr = 8\sqrt{\pi^3}\sigma\rho a^3 C, \tag{9.120}$$

where ρ is the constant density of the medium and σ is its specific heat. The final result in Eq. (9.120) is obtained by first making a change of variable from r to $\xi = r/2a\sqrt{t}$, obtaining

$$\int_0^\infty e^{-r^2/4a^2t} r^2 dr = (2a\sqrt{t})^3 \int_0^\infty e^{-\xi^2} \xi^2 d\xi,$$

then evaluating the ξ integral via an integration by parts:

$$\int_0^\infty e^{-\xi^2} \xi^2 d\xi = -\frac{\xi}{2} e^{-\xi^2}\Big|_0^\infty + \frac{1}{2}\int_0^\infty e^{-\xi^2} d\xi = \frac{\sqrt{\pi}}{4}.$$

The temperature, as given by Eq. (9.119) at any moment, i.e., at fixed t, is a Gaussian distribution that flattens out as time increases, because its width is proportional to $\sqrt{t}$. As a function of time the temperature at any fixed point is proportional to $t^{-3/2}e^{-T/t}$, with $T \equiv r^2/4a^2$. This functional form shows that the temperature rises from zero to a maximum and then falls off to zero again for large times. To find the maximum, we set

$$\frac{d}{dt}\left(t^{-3/2}e^{-T/t}\right) = t^{-5/2}e^{-T/t}\left(\frac{T}{t} - \frac{3}{2}\right) = 0, \qquad (9.121)$$

from which we find $t_{\max} = 2T/3 = r^2/6a^2$. The temperature maximum arrives at later times at larger distances from the origin. ■

In the case of **cylindrical symmetry** (in the plane $z = 0$ in plane polar coordinates $\rho = \sqrt{x^2 + y^2}, \varphi$), we look for a temperature $\psi = u(\rho, t)$ that then satisfies the ODE (using Eq. (2.35) in the diffusion equation)

$$\frac{\partial u}{\partial t} = a^2\left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial u}{\partial \rho}\right), \qquad (9.122)$$

which is the planar analog of Eq. (9.118). This ODE also has solutions with the functional dependence $\rho/\sqrt{t} \equiv r$. Upon substituting

$$u = v\left(\frac{\rho}{\sqrt{t}}\right), \quad \frac{\partial u}{\partial t} = -\frac{\rho v'}{2t^{3/2}}, \quad \frac{\partial u}{\partial \rho} = \frac{v'}{\sqrt{t}}, \quad \frac{\partial^2 u}{\partial \rho^2} = \frac{v'}{t} \qquad (9.123)$$

into Eq. (9.122) with the notation $v' \equiv dv/dr$, we find the ODE

$$a^2 v'' + \left(\frac{a^2}{r} + \frac{r}{2}\right) v' = 0. \qquad (9.124)$$

This is a first-order ODE for v', which we can integrate when we separate the variables v and r as

$$\frac{v''}{v'} = -\left(\frac{1}{r} + \frac{r}{2a^2}\right). \qquad (9.125)$$

This yields

$$v(r) = \frac{C}{r} e^{-r^2/4a^2} = C\frac{\sqrt{t}}{\rho} e^{-\rho^2/4a^2t}. \qquad (9.126)$$

This special solution for cylindrical symmetry can be similarly generalized and adapted to boundary conditions, as for the spherical case. Finally, the z-dependence can be factored in, because z separates from the plane polar radial variable ρ.

Exercises

9.7.1 For a homogeneous spherical solid with constant thermal diffusivity, K, and no heat sources, the equation of heat conduction becomes

$$\frac{\partial T(r,t)}{\partial t} = K\nabla^2 T(r,t).$$

Assume a solution of the form

$$T = R(r)T(t)$$

and separate variables. Show that the radial equation may take on the standard form

$$r^2\frac{d^2R}{dr^2} + 2r\frac{dR}{dr} + \alpha^2 r^2 R = 0,$$

and that $\sin\alpha r/r$ and $\cos\alpha r/r$ are its solutions.

9.7.2 Separate variables in the thermal diffusion equation of Exercise 9.7.1 in circular cylindrical coordinates. Assume that you can neglect end effects and take $T = T(\rho, t)$.

9.7.3 Solve the PDE

$$\frac{\partial\psi}{\partial t} = a^2\frac{\partial^2\psi}{\partial x^2},$$

to obtain $\psi(x,t)$ for a rod of infinite extent (in both the $+x$ and $-x$ directions), with a heat pulse at time $t = 0$ that corresponds to $\psi_0(x) = A\delta(x)$.

9.7.4 Solve the same PDE as in Exercise 9.7.3 for a rod of length L, with position on the rod given by the variable x, with the two ends of the rod at $x = 0$ and $x = L$ kept (at all times t) at the respective temperatures $T = 1$ and $T = 0$, and with the rod initially at $T(x) = 0$, for $0 < x \leq L$.

9.8 Summary

This chapter has provided an overview of methods for the solution of first- and second-order linear PDEs, with emphasis on homogeneous second-order PDEs subject to boundary conditions that either determine unique solutions or define eigenvalue problems. We found that the usual boundary conditions are identified as of Dirichlet type (solution specified on boundary), Neumann type (normal derivative of solution specified on boundary), or Cauchy type (both solution and its normal derivative specified). Applicable types of boundary conditions depend on the classification of the PDE; second-order PDEs are classified as hyperbolic (e.g., wave equation), elliptic (e.g., Laplace equation), or parabolic (e.g., heat/diffusion equation).

The method of widest applicability to the solution of PDEs is the method of separation of variables, which, when effective, reduces a PDE to a set of ODEs. The chapter has presented a very small number of complete PDE solutions to illustrate the technique. A wider variety of examples only becomes possible when we are prepared to exploit the properties of the special functions that are the solutions of various ODEs, and, as a result, fuller illustration of PDE solutions will be provided in the chapters that discuss these special functions. We point out, in particular, that general PDEs with spherical symmetry all have the same angular solutions, known as **spherical harmonics**. These, and the functions from which they are constructed (Legendre polynomials and associated Legendre functions), are the subject matter of Chapters 15 and 16. Some spherically symmetric problems have radial solutions that can be identified as **spherical Bessel functions**; these are treated in the Bessel function chapter (Chapter 14).

PDE problems with cylindrical symmetry usually involve Bessel functions, often in ways more complex than in the examples of the present chapter. Further illustrations appear in Chapter 14.

This chapter has not attempted to discuss methods for the solution of inhomogeneous PDEs. That topic deserves its own chapter, and will be developed in Chapter 10.

Finally, we repeat an earlier observation: Fourier expansions (Chapter 19) and integral transforms (Chapter 20) can also have a role in the solution of PDEs, and applications of these techniques to PDEs are included in the appropriate chapters of this book.

Additional Readings

Bateman, H., *Partial Differential Equations of Mathematical Physics*. New York: Dover (1944), 1st ed. (1932). A wealth of applications of various partial differential equations in classical physics. Excellent examples of the use of different coordinate systems, including ellipsoidal, paraboloidal, toroidal coordinates, and so on.

Cohen, H., *Mathematics for Scientists and Engineers*. Englewood Cliffs, NJ: Prentice-Hall (1992).

Folland, G. B., *Introduction to Partial Differential Equations*, 2nd ed. Princeton, NJ: Princeton University Press (1995).

Guckenheimer, J., P. Holmes, and F. John, *Nonlinear Oscillations, Dynamical Systems and Bifurcations of Vector Fields*, revised ed. New York: Springer-Verlag (1990).

Gustafson, K. E., *Partial Differential Equations and Hilbert Space Methods*, 2nd ed., New York: Wiley (1987), reprinting Dover (1998).

Margenau, H., and G. M. Murphy, *The Mathematics of Physics and Chemistry*, 2nd ed. Princeton, NJ: Van Nostrand (1956). Chapter 5 covers curvilinear coordinates and 13 specific coordinate systems.

Morse, P. M., and H. Feshbach, *Methods of Theoretical Physics*. New York: McGraw-Hill (1953). Chapter 5 includes a description of several different coordinate systems. Note that Morse and Feshbach are not above using left-handed coordinate systems even for Cartesian coordinates. Elsewhere in this excellent (and difficult) book are many examples of the use of the various coordinate systems in solving physical problems. Chapter 6 discusses characteristics in detail.